

Highlights

Local skill, unstable portability: sign reversal in pre-fire thermal predictors across Mediterranean wildfire regions, and the diagnostics that do not order it

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- Pre-fire thermal predictors add +0.06 to +0.15 AUC within all five fire regions
- The same block adds +0.004 AUC across 20 transfer directions, interval spans zero
- Highest niche-overlap pair fails both ways; lowest transfers both ways (points)
- Of 20 diagnostics, only conditional direction agreement tracks transfer (rho 0.84)
- Label-blind adaptation compresses transfer toward chance; only target labels help

Local skill, unstable portability: sign reversal in pre-fire thermal predictors across Mediterranean wildfire regions, and the diagnostics that do not order it

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Abstract

Pre-fire thermal dryness separates a fire year from a normal year. Models built on it are rarely tested outside the region they were fitted in. Five Mediterranean wildfire regions were analysed here, on about 500 m cells, with MCD64A1 labels and spatially blocked validation. Six pre-fire thermal predictors were added to a terrain, fuel and greenness baseline.

The local skill does not travel, and neither does the baseline. Within-region ROC-AUC rose by +0.056 to +0.153 in every region. Across twenty ordered transfer directions the same block contributed +0.004 [−0.028, +0.036]. That is not distinguishable from zero, and its sign changes from pair to pair. The static baseline transferred no better, at 0.537 against 0.541. Two predictors reverse their association between regions: elevation and the LST anomaly. Removing them cost −0.081 of within-region skill, supported in every region and mostly due to elevation. It changed transfer by +0.014 [−0.017, +0.045]. Both figures are post-selection estimates. A local cost is measured. No compensating transfer gain is.

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The failure is invisible to the diagnostics that can be run before deployment. Twenty candidate diagnostics were rank-correlated against observed transfer. Only two had intervals excluding zero, and both were conditional. The stronger one measures agreement in the sign of each predictor’s association (Spearman $\rho = 0.84$ over sixteen directions). No marginal measure was shown to order the matrix, including predictor-space dissimilarity. At the point estimates, the pair with the highest burned-niche overlap failed in both directions, while the lowest transferred in both. Signed associations need labels on both sides. The family that works is therefore the one a practitioner does not have.

The mechanism is a reversal of sign. Predictors do not weaken across regions. They reverse the direction of their association with burning. A distance in predictor space cannot see that. The same reversal appears inside one study area, between two fires eleven months apart on an identical grid. That arm rests on one fire and eleven positive-carrying blocks, so it corroborates rather than establishes.

Label-free alignment pushed fourteen of twenty directions towards chance rather than repairing them. Transfer skill has to be measured, not inferred from similarity. The price is target labels: thirty-two labelled 5 km blocks recovered 85 to 89 % of the target’s matched ceiling in three of six directions.

Keywords: wildfire, model transferability, area of applicability, land surface temperature, domain adaptation, concept shift

1. Introduction

Wildfire is a defining disturbance of Mediterranean-basin landscapes, and a changing climate is reshaping where and how it burns [27]. The pressure to anticipate *where* fire will occur has produced a large and still-growing literature on fire susceptibility mapping. It has recently been synthesised from both a methodological and an application perspective [39, 13]. The dominant pattern is well established. Geospatial predictors are assembled over a study region and paired with a historical record of burned area. A supervised classifier is then fitted, most often a random forest [4]. The resulting susceptibility surface is published with a cross-validated AUC in the 0.85 to 0.95 range.

This paper is about a property that the pattern does not report. Predictors that describe the *state* of a surface in a particular season are more informative about that season than predictors that describe a place. They are also, as we show, not portable: the relationship they carry is reparameterised locally, existing everywhere but pointing in different directions in different places. The dynamic block is worth +0.056 to +0.153 ROC-AUC inside every region and contributes +0.004 between them, an estimate whose interval spans zero, with a sign that is a property of the source-target pair rather than of the block.

The obvious reading of that pair of numbers is a trade-off, and our own earlier framing took it. The data do not support it. The static baseline transfers at a mean of 0.537 against the thermal model’s 0.541, so the predictor class that ought to travel does not travel either, and removing the reversing predictors returns +0.014 of transfer on an interval that also spans zero. Two

nulls on the portability axis are not an exchange. What is measured is a local gain that does not export, and a local cost when the reversing predictors are removed, with no compensating transfer gain on either arm.

1.1. Portability goes unmeasured

30 A susceptibility model’s reported skill is almost always an estimate of *within-region* performance. Held-out folds come from the same study area and season, often from the same fire event, and where folds are drawn at random over cells, spatial autocorrelation inflates the estimate further. The problem is documented across ecological modelling [31, 29] and is addressed
35 by spatially blocked cross-validation [37, 23]. Blocking makes the within-region estimate honest. It says nothing about whether the fitted relationship holds elsewhere.

Because only that side of the ledger is reported, the trade-off is structurally invisible. A predictor block is adopted on the strength of the in-
40 crement it delivers inside its training footprint, and the portability it consumes is never entered as a debit, even though any regional product built from locally trained models implicitly promises generalisation beyond that footprint. For fire specifically the transfer question is asked far less often than in adjacent fields. Meteorological fire-danger indices are known not
45 to port cleanly between fire environments [30], and the two recent studies that test model transfer systematically, across Mediterranean countries [7] and US counties [15], both report that it largely succeeds between similar regions. Both transfer models whose dominant predictors are *spatially stationary*: terrain, human modification, night-time lights, population density,
50 long-term climatologies. Each describes a place rather than a season, which is

the class our thesis expects to travel well. Species distribution modelling has examined transferability far more systematically [43], and there geographic and environmental similarity do *not* reliably predict transfer success [38, 32].

1.2. Pre-fire thermal dryness is the natural test case

55 The dynamic class used here is pre-fire thermal dryness, and it is the class most plausibly *expected* to transfer. Standard susceptibility predictors are terrain, fuel type, greenness, long-term climatology and proximity to roads and settlements, and most are static or near-static over the timescale at which fire danger varies. Such a set explains poorly why a particular summer burned and the preceding one did not. What changes between those
60 summers is the state of the surface: fuel dryness, moisture stress, and how anomalously hot the land surface has become relative to its own baseline. Satellite thermal observation gives direct access to part of that state through fuel moisture content [44], and pairing surface temperature with a vegetation index was established as a live-fuel-moisture estimator for fire-danger
65 rating by Chuvieco et al. [6]. The Temperature-Vegetation Dryness Index [33] formalises that feature space into an internally normalised measure. In principle it should be less exposed to absolute-temperature offsets between regions than raw land surface temperature. That theoretical portability advantage is tested here, and it is not found: the two internally normalised
70 channels transfer no better than the four absolute ones, at means of 0.544 and 0.548 over the twenty directions (Appendix A(f)).

That pre-fire thermal state carries genuine information about subsequent fire is established [17, 19, 18, 8]. What has not been established is whether
75 the predictive *skill* of a classifier trained on it in one fire region survives

strict, label-free application to another. The physics linking moisture stress to combustion is universal, so portability should be most expected for this class, which is what makes it diagnostic: a loss here cannot be dismissed as a peculiarity of a locally defined covariate.

80 *1.3. Why the loss is invisible to the diagnostics in use*

A deficit on transfer can arise two ways. Under **covariate shift** the marginal distribution of the predictors differs while the predictor-response relationship is preserved, which is in principle correctable without target labels, by per-region standardisation or covariance alignment such as CORAL
85 [35], an approach with an established remote-sensing literature [36]. Under **concept shift** the conditional relationship itself changes: a predictor positively associated with burning in one region is negatively associated with it in another. No label-free realignment of input distributions can repair a sign reversal, because the information needed to detect it is exactly the
90 information being withheld. The taxonomy is canonical [25].

This distinction is what makes the failure hard to anticipate. The diagnostics the field relies on to decide whether a model may be applied to new ground, above all area-of-applicability measures built on dissimilarity in predictor space [21, 22, 16], are marginal by construction. A target region
95 can sit well inside the training data’s predictor envelope while the relationship between those predictors and burning points the other way. This paper measures both sides on the same pairs.

1.4. Contributions

Three findings carry this paper. Each is stated at the strength its interval
100 supports, and the nearest prior work is named inside the claim.

Contribution 1. Local skill and portability, measured separately and per direction. The thermal block is worth $+0.056$ to $+0.153$ ROC-AUC inside every one of five regions, with every bootstrap interval above zero and the result holding in both analysis populations. Across the twenty
105 ordered transfer directions it contributes $+0.004$. That interval spans zero under all four between-direction resampling units we computed, and the sign varies by pair: paired deltas run from -0.148 to $+0.132$, twelve positive and eight negative. The baseline arm bounds the reading. It transfers at a mean of 0.537 , against the thermal model’s 0.541 . The static predictor class is not
110 the portable one here either. What the matrix shows is that **nothing in this feature space travels**, not that dynamic state is uniquely fragile. Feature removal measures the local cost. Dropping the two reversing predictors, elevation and the LST anomaly, costs -0.081 of mean within-region AUC, supported in every region. Roughly three quarters of that is elevation, a *baseline*
115 terrain variable. Mean transfer changes by $+0.014$, whose interval also spans zero. Both are post-selection estimates (Section 4.6b). A local cost is measured. No compensating transfer gain is. The within-region half is not novel, comparable results existing for these landscapes [1, 12]; the paired contrast against portability is. Dimarco et al. [7] transfer a predominantly stationary
120 predictor set successfully across a comparable Mediterranean design, though their response variable is ignition rather than burned area (Section 5.7).

Contribution 2. The loss is invisible to the diagnostics that

could be run before deployment, and visible to one that cannot.

Twenty candidate diagnostics from four families are evaluated against observed transfer under one bootstrap framework. Area-of-applicability-style predictor-space dissimilarity [21, 22, 16], climatic and geographic distance, learned domain separability, niche-overlap statistics and burn-pattern regime distances were none of them shown to order the matrix; the domain classifier is at ceiling, with $AUC \geq 0.96$ for every pair. Only two diagnostics have intervals excluding zero. Both measure conditional direction agreement, that is, whether each predictor’s signed association with burning points the same way in both regions (Spearman $\rho = +0.84$ and $+0.81$ over sixteen directions from eight pairs). Both are computed on a data-selected subset of predictors, and their all-nine-feature counterparts span zero, so the result rests on that selection. Equalising the families onto a common twelve directions leaves the ordering unchanged, so it is not an artefact of unequal samples. Signed associations need burned labels on both sides, so this is a mechanism diagnosis rather than a pre-deployment screen, and the marginal family, the only one runnable before deployment, is the one that fails. The same point survives without any ranking, at the point estimates: the pair with the highest burned-niche overlap fails in both directions while the lowest transfers in both. That contradicts an expectation the fire literature carries [7, 15] and agrees with species distribution modelling [38, 32]. Shift decomposition in applied remote sensing is not itself new [11]; the fire application and the head-to-head of four diagnostic families on the same pairs are.

Contribution 3. The mechanism is a reversal of sign, and it survives holding geography fixed. Predictors do not merely weaken across

regions, they reverse the direction of their association with burning, which is why a distance in predictor space cannot see the failure. The sharpest reversal is elevation, whose association points opposite ways in Manavgat and in Bejís and Muğla with disjoint bootstrap intervals. The same reversal appears inside a single study area, between two fires eleven months apart on an identical grid, where season and population also differ but place does not (Section 4.8). That arm rests on one fire and eleven positive-carrying 5 km blocks, below the floor this design sets for itself, so it corroborates the mechanism rather than establishing it.

Two consequences follow. Both are reported as supporting results, not leading ones. First, label-free alignment by standardisation and covariance alignment [35] does not repair transfer. It compresses transfer towards chance in fourteen of twenty directions. We believe this is the first application of covariance alignment to fire susceptibility. Second, the residual gap is conditional, so the resource that closes it is target labels. That price is measurable. Thirty-two labelled 5 km blocks recover 85 to 89 % of the target’s matched ceiling in three of six directions, 51 to 57 % in two more, and 30 % in the sixth.

Alongside these, a leakage-audited, spatially blocked evaluation and transfer protocol is released with code, configuration and frozen outputs, so that most of this result can be re-run rather than taken on trust. One component is an exception and is named in the data-and-code availability statement: the reproduction-check code is not in the released repository at the commit of record. The few-shot export, previously also listed as an exception, was regenerated at commit 6d7a6a71 and its validator now passes on every check.

This matters given evidence that wildfire transfer conclusions are sensitive to evaluation design and task formulation [42].

175 A companion paper treats the observational layer beneath this one, where preprocessing decisions taken before any model is fitted are shown to carry their own uncertainty budget.

Section 2 reviews the relevant literature. Section 3 describes the study regions, the data, and the modelling, cross-validation, transfer and adapta-
180 tion protocol. Section 4 reports results. Section 5 discusses them, including limitations. Section 6 concludes.

2. Related work

2.1. Fire susceptibility mapping, and what it reports

Fire susceptibility mapping has been synthesised recently from both method-
185 ological and application perspectives [39, 13]. The dominant pattern is stable: geospatial predictors are assembled over a study region, paired with a historical burned-area record, and fitted with a supervised classifier, most often a random forest [4], and the resulting surface is published with a cross-validated AUC in the 0.85 to 0.95 range. That figure is a within-region
190 estimate. Comparable within-region results already exist for the landscapes studied here [1, 12].

2.2. Pre-fire thermal dryness

Satellite thermal observation gives repeated access to surface state through fuel moisture content [44]. Pairing land surface temperature with a vegeta-
195 tion index was established as a live-fuel-moisture estimator for fire-danger

rating by Chuvieco et al. [6], and the Temperature-Vegetation Dryness Index [33] formalises that feature space into an internally normalised measure, which is why it is often expected to travel between regions better than raw temperature. That pre-fire thermal state carries information about subsequent fire is established [17, 19, 18]. Gelabert et al. [8] found dead fine fuel moisture and its anomalies to be the most influential predictor of human-caused ignition likelihood across Europe, testing generalisation by pooled fitting with per-site evaluation. What has not been tested is whether a classifier built on this class survives strict, label-free application to a region it never saw.

2.3. Spatial validation and transferability

Random cross-validation over spatially autocorrelated cells inflates skill estimates [31, 29], and spatially blocked designs are the standard remedy [37, 23]. The remedy is not uncontested: blocked schemes have been argued to introduce pessimistic bias of their own [40, 24, 5]. This paper uses spatial blocking throughout and reports every result at three block sizes, so that a reader can see which verdicts depend on the choice.

Blocking makes a within-region estimate honest but says nothing about another region. For that, the field’s instrument is the area of applicability and related predictor-space dissimilarity measures [21, 22, 16], which ask whether a target’s predictor values fall inside the training data’s envelope. Species distribution modelling has examined transferability far more systematically than fire has [43], and two studies there report that geographic and environmental similarity do not reliably predict transfer success [38, 32], and the transferability of satellite-derived empirical models is an active question

in fire remote sensing itself [20].

2.4. Dataset shift and label-free adaptation

The canonical taxonomy separates covariate shift, where the predictor distribution moves but the predictor-response relationship holds, from concept shift, where the relationship itself changes [25]. Covariate shift is in principle correctable without target labels, by per-region standardisation or by covariance alignment such as CORAL [35], and domain adaptation has an established remote-sensing literature [36, 28]. Concept shift is not: no realignment of input distributions can repair a reversal in the sign of an association, because detecting it requires the labels being withheld. Shift decomposition in applied remote sensing is not itself new [11]. We are not aware of a prior application of covariance alignment to fire susceptibility, fire occurrence or burned-area prediction.

2.5. Cross-region generalisation of fire models

Few studies test fire-model transfer directly. Podschwit et al. [30] give the long-standing reference point that meteorologically derived danger indices do not port cleanly between fire environments. WildfireGenome [15] runs a leave-one-county-out matrix across seven US counties and reports strong within-county performance with highly variable off-diagonal transfer, similar pairs transferring well and dissimilar pairs collapsing; its label, however, is a principal-component composite of hazard *indicators* rather than observed burned area, and it applies no adaptation. Xu et al. [42] argue that wildfire transfer conclusions depend strongly on evaluation design, a caution we address by fixing the protocol in a project log before the diagnostics were

245 computed and by reporting every sensitivity axis; that log is not a formal pre-registration, and Section 3.11 states what was fixed and when. Kondylatos et al. [14] provide Mesogeos, a 1 km Mediterranean datacube for data-driven wildfire modelling.

The nearest neighbour, and the contrast the present paper draws.

250 Dimarco et al. [7] is the closest Mediterranean analogue: 500 m predictors harmonised across four countries, tree ensembles under spatial cross-validation, transfer tested both leave-one-country-out and as a full 4 x 4 matrix, with every transfer exceeding AUC 0.80 and bioclimatically similar countries scoring higher. They apply no domain adaptation.¹

255 Much of the design is shared — Mediterranean regions, 500 m cells, MCD64A1-derived targets, tree ensembles, spatially aware validation, an explicit transfer matrix — and two things differ. **The predictor class:** their model rests on attributes of a place, all spatially stationary, ours on the state of a surface in one season. **The response variable:** theirs is human-driven
260 ignition, ours is burned area, and those are different quantities with different dominant controls. Read together the two results suggest that the relationship between domain similarity and transfer success may be predictor-class dependent, with Vesk et al. [38] and Rousseau and Betts [32] aligning from species distribution modelling while Dimarco et al. and WildfireGenome run
265 against it. We present that as a live disagreement, and Section 5.7 states why our own data do not settle it.

Methods, where no land surface temperature or TVDI variable appears

¹Their Results text refers to "LST anomalies", which is inconsistent with their own

and the only temperature predictor is a static ERA5-Land seasonal climatology [26]. We follow their Methods, which we take to be authoritative, and
 270 note the discrepancy so that the comparison is transparent. Every statement about their work was checked against the full text, read 2026-08-08.

3. Data and methods

3.1. Study regions and temporal windows

Five Mediterranean wildfire regions are analysed: Manavgat 2021 and
 275 Muğla 2021 in Türkiye, Bejís 2022 in Spain, North Evia 2021 in Greece and Montiferru 2021 in Sardinia. Each is a place-based rectangular area of interest in EPSG:4326, defined from place coverage rather than from a fire perimeter, and not tuned on burned prevalence, gate outcome or any model metric. One AOI choice was label-informed and is stated as such. The North
 280 Evia box was extended after the legacy box was found to carry an atypically high burned prevalence. The extended geometry was then defined from place anchors, and the legacy variant is kept as a sensitivity arm (Appendix A). The AOIs are deliberately not clipped to fire perimeters, so that unburned cells around each fire form the negative class rather than being excluded by
 285 construction. A sixth region, Kozan 2023, is carried as a negative control and is excluded from all modelling by the gate of Section 3.3.

Each region has two non-overlapping windows. The **predictor window** closes the day before the **label window** opens, so no predictor observation can post-date the first labelled burning. Window lengths are matched within
 290 a region and vary between them with the event: 56 to 60 days for predictors and 35 to 49 days for labels. A four-year baseline of window-symmetric

composites precedes each predictor window and supplies the climatological reference for the anomaly channels. Region identifiers, bounding boxes, window dates and baseline years are registered in `core/regions.py` and are reproduced in Table 1.

3.2. Burned-area label and the ~ 500 m analysis grid

Labels come from MODIS MCD64A1 Collection 6.1 [9] retrieved through Google Earth Engine [10], whose omission and commission characteristics [3] bound every model here. The analysis grid is **reconstructed rather than native**: the pipeline’s 30 m EPSG:4326 reference grid is partitioned into 17 x 17 blocks, giving a nominal 510 m cell that approximates rather than reproduces the MODIS cell, and is square in degrees but not on the ground. A cell’s burn date is the mode of its positive sub-pixel day-of-year values, tested against the region’s label window; the label never affects a cell’s eligibility for modelling. Two safeguards are recorded rather than assumed: cells burning inside a region’s own predictor window are removed, which ran for three of five regions, and burning in earlier years is screened for none. Appendix C.1 gives the full specification, including what follows from the grid’s shape.

3.3. Burned-landcover admissibility gate

Before any modelling each region passes a gate asking what fraction of its burned cells is dominated by natural vegetation, using ESA WorldCover classes on the same cells. A region is admitted at 0.50 with at least 30 burned cells, and rejected as a cropland-dominated control at 0.50 cropland. The purpose is to separate burned area produced by natural-fuel combustion

from post-harvest stubble burning, which MCD64A1 does not distinguish. Verdicts are in Section 4.1; the full rule is in Appendix C.1.

3.4. Predictor variables

Ten predictors are used, four baseline and six thermal, all summarised
320 per cell over the predictor window. All optical and thermal predictors come from Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1_L2), quality-screened per pixel from the QA_PIXEL band with the water bit deliberately preserved; the coarse-resolution thermal input is MODIS/061/MOD11A1.

- **NDVI**, the predictor-window median of Landsat surface reflectance, is
325 the baseline’s one time-varying member.
- **Elevation** and **slope** come from the Copernicus DEM GLO-30.
- **Land cover** is ESA WorldCover v200 [45], entering as the dominant class code.
- **Current LST**, the predictor-window median of Landsat surface tem-
330 perature in °C.
- **LST anomaly**, a z-score of the current-window LST median against the four baseline years, set to no-data where the baseline standard deviation is below 1.0 °C or observations are too few.
- **TVDI** and **TVDI difference**. The Temperature-Vegetation Dryness
335 Index [33] is computed in the LST-NDVI feature space, with wet and dry edges taken as the 2nd and 98th LST percentiles within each of twenty NDVI bins and the index clamped to [0, 1]. The edges are

percentiles of the LST values a given scene contains, so they are fitted per AOI and per window, and a TVDI of 0.5 denotes a different physical moisture state in each region. The companion paper reports what that scene dependence does and does not explain. **TVDI difference** is the raw anomaly of that index against the same four window-symmetric baseline years used for the LST anomaly: `tvdi_difference = current_tvdi - mean(baseline_tvdi)`, in index units rather than standard deviations, which is why it is reported alongside the z-scored channel rather than in place of it.

- **Downscaled LST and fused LST.** A MODIS-to-Landsat downscaling model is trained on the predictor window and applied to the full 30 m grid. The fused product equals observed Landsat LST wherever that is valid, and the downscaled surface only where it is not. Gap-filling therefore never replaces or blends a valid observation. The gap-filled share is 0.11 % to 9.70 % by region. The downscaler’s own inputs include coordinates, which is the one route by which a coordinate-derived surface re-enters a feature set from which Section 3.13 excludes coordinates; the companion paper measures what the increment is worth without these two channels.

3.5. Cell aggregation, validity and analysis populations

Cell-level values are means over the ~ 510 m cell, computed from valid 30 m pixels only, with the valid fraction recorded. A cell is `valid_for_modeling` when it meets a 30 % valid-pixel floor and no thermal channel is missing. Two populations are then defined. The **primary** population

is natural vegetation, cells whose combined tree, shrub and grass fraction reaches 0.50, which excludes cropland from every burnable mask. The **secondary** population is all valid cells, reported throughout as a sensitivity.

365 Every headline result is given for both.

3.6. Feature sets and classifier

Two nested feature sets are compared. The **baseline** set is NDVI, elevation, slope and land cover. The **thermal** set is the baseline plus the six thermal channels. Land cover is one-hot encoded and numeric features
370 are standardised inside the fitting pipeline. The classifier is a random forest [4] with 300 trees, unlimited depth, `min_samples_leaf = 3`, balanced class weights and `random_state = 42`, identical for every region, population, feature set and transfer direction, so that no comparison in this paper is confounded by a model choice.

375 3.7. Spatial-block cross-validation and bootstrap uncertainty

Cross-validation is spatially blocked. Blocks are squares of the analysis grid formed as `row_500m // B by col_500m // B`, and folds are drawn over whole blocks, so no block is split between training and test. Five folds are used. The block size B is reported at 2, 10 and 20 cells, about 1, 5 and 10
380 km, for the within-region analysis.

Uncertainty is a spatial-block bootstrap: blocks are resampled with replacement, 1000 replicates, seed 42, with 2.5 and 97.5 percentiles as the interval. Because it is blocks that are resampled, what bounds an interval's reliability is the number of blocks carrying at least one burned cell, and those

385 counts are reported alongside the intervals. Where a verdict rests on too few
such blocks it is stated as indicative rather than as an interval.

3.8. Cross-region transfer protocol

For each ordered pair of regions the model is fitted on the source pop-
ulation and applied to the target population with **no target label used**
390 **at any point**: no refitting, no threshold selection, no calibration. Both
the baseline and the thermal feature set are transferred, so that the thermal
block’s contribution is a paired difference within a direction rather than a
comparison across directions. All twenty ordered directions among the five
regions are computed.

395 3.9. Label-blind domain adaptation

Two label-free remedies are tested on every direction, in both feature sets.

Region-wise z-score. Each region’s numeric features are standardised
using its own statistics, source statistics from source data and target statistics
from target data, never pooled. The classifier is refitted on the z-scored
400 source and applied to the z-scored target. This removes first- and second-
order marginal offsets.

CORAL after region-wise z-score. Starting from the z-scored fea-
tures, the source covariance is aligned to the target covariance by the stan-
dard whitening-recolouring map [35], with regularisation $\lambda = 10^{-5}$ and eigen-
405 values floored at 10^{-12} . Critically **the transform is applied to the source**
only; the target is left exactly as it is. The classifier is refitted on the aligned
source. Neither variant sees a target label, and both are verified label-blind
at run time.

λ sensitivity was assessed over nine values from 0 to 10^{-1} on four di-
 410 rections in both feature families, moving transfer AUC by at most 0.014 in
 any direction and 0.008 within the thermal family. The canonical $\lambda = 1$ lies
 outside the released sweep; the companion paper reports it.

3.10. *Transfer-gap decomposition and the concept-shift diagnostic*

For each direction the gap between the target’s own within-region skill
 415 and the raw transfer result is split in two. One part is what the best label-free
 adaptation recovers, and the other is what it does not. The recovered fraction
 is defined as (adapted – raw) / (within – raw), signed and unclipped, and its
 interval comes from the same paired bootstrap. The recovered part bounds
 what covariate-level correction can achieve. The remainder is the conditional
 420 residual.

The mechanism is diagnosed by **signed univariate association**. For
 each numeric predictor the raw ROC-AUC of that predictor against **burned**
 is computed in each region and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a
 value below 0.5 is read as a direction rather than as weakness. A reversal is
 425 called bootstrap-supported only when the two regions’ intervals are disjoint,
 under the same 10-cell spatial-block bootstrap.

3.11. *Transferability diagnostics versus transfer*

Twenty candidate diagnostics from four families are computed for every
 region pair and rank-correlated (Spearman) against observed raw transfer
 430 AUC, under one bootstrap framework that resamples unordered region pairs
 with both of their ordered directions travelling together. The families are

marginal predictor-space measures $P(x)$, including area-of-applicability dissimilarity, climatic and geographic distance and a learned domain classifier; burned-niche overlap $P(x|y=1)$; burn-pattern regime distance $P(y)$; and conditional direction agreement $P(y|x)$. No family-wise error control is claimed, and the number of variants computed within each family is reported with the result. Appendix C.1 gives the full specification of each diagnostic and of the bootstrap.

3.12. Interventions

Pooled multi-region training. Leave-one-region-out: the model is trained on the pooled primary populations of the other four regions and evaluated on the held-out region, with folds blocked as before. This asks whether pooling recovers what single-source transfer loses.

Removal of direction-reversing features. The two predictors whose signed association reverses between regions with bootstrap support are **elevation_mean** (Manavgat against Bejís and against Muğla) and **lst_anomaly_mean** (Bejís against Evia). They are dropped and everything is refitted, within-region and across every direction, and the two are also dropped singly so the cost can be attributed. This measures what the removal costs locally and what it returns on transfer on the same footing. Note that the two features are selected by the same reversal analysis against which the result is then read, so both quantities are post-selection estimates and no correction for that selection is applied.

3.13. Leakage control

455 An explicit forbidden-column set is enforced at every model fit. Coordinates (`lon`, `lat`, `row`, `col` and their normalised forms), every burn-date and label-provenance column, and the agreement fraction are excluded from all feature sets, and the enforcement runs as an assertion rather than a convention. The natural-vegetation mask is used only to define the population,
460 never as a predictor. Spatial blocking prevents a cell from sharing a fold with its own neighbours.

Reproducibility. All randomness uses seed 42 and the bootstrap uses 1000 replicates throughout. The transfer and adaptation analysis runs in an environment separate from the upstream pipeline’s. Every within-region
465 model was therefore refitted there and compared against the frozen upstream output, and the independently implemented adaptation was compared against the pipeline’s own. The within-region comparisons agree exactly, and the twenty transfer directions agree to within 1.6×10^{-7} . Random-forest fits are not bit-identical across library versions. All numbers here were
470 therefore produced under scikit-learn 1.9.0, or verified against it. If that version is not fixed, cross-region point estimates carry an implementation tolerance of about ± 0.02 to 0.03. The companion paper reports the version sensitivity and the reproduction check in full.

Sensitivity analyses. Every headline result is repeated across two analysis
475 populations, three spatial-block sizes, the CORAL regularisation sweep and both feature sets. Where a conclusion depends on one of those choices, the dependence is reported rather than resolved by choosing the favourable setting.

3.14. *Same-geography event-to-event comparison (Mugla 2021 versus 2022)*

480 Mugla admits a comparison in which place is held fixed and the event varies: a second fire burned inside the identical AOI, on the identical analysis grid, eleven months after the first. Signed univariate AUCs are computed for both arms under the same 10-cell spatial-block bootstrap used elsewhere. Season, year and population all differ between the arms, since the 2022 arm
485 is defined by removing the 2021 scar, and Section 4.8 reports what that costs. Appendix C.2 gives the window dates, the population construction and the asymmetry audit.

4. Results

4.1. *Study regions and the admissibility gate*

490 All five candidate regions pass the burned-landcover gate as wildfire candidates and the negative control fails it as intended. Kozan 2023 returns a natural-vegetation fraction of 0.017 against the 0.50 threshold and is excluded from all modelling. The separation is not marginal: the five admitted regions carry fractions of 0.723 to 0.991, so the threshold falls in an empty
495 interval rather than between neighbouring cases. That is consistent with the gate separating burned area produced by natural-fuel combustion from post-harvest stubble burning, which MCD64A1 does not distinguish, though one negative control cannot establish it.

North Evia is analysed on an extended AOI. Relative to the legacy box,
500 the extended box leaves the burned scar essentially unchanged while cutting overall prevalence from 0.361 to 0.122 and TSG prevalence from 0.676 to 0.287. The effect of that choice on transfer is reported in Appendix A(a).

Table 1: Region summary (recap of Table 1; final numbering at assembly). Counts from each region’s Step 8A dataset statistics; gate fractions from each region’s burned-landcover gate output. TSG = the primary natural-vegetation population. The TSG columns use the canonical modelled population, **burnable_tree_shrub_grass and valid_for_modeling == True**, which is the population every model in this paper was fitted and scored on.

Region	Total cells	Valid cells	Burned	Prevalence (all valid)	TSG cells	Burned in TSG	TSG prevalence	Burned natural- veg frac- tion	Gate verdict
Manavgat 2021	24,150	24,087	796	0.033	20,511	784	0.038	0.984	pass
Bejis 2022	15,759	15,759	1,103	0.070	15,190	1,100	0.072	0.991	pass
Muğla 2021	73,098	73,045	3,026	0.041	41,730	2,911	0.070	0.958	pass
North Evia 2021 (ex- tended)	22,925	22,906	2,788	0.122	9,298	2,664	0.287	0.945	pass
Monticelli 2021	3,234	3,173	697	0.220	2,544	539	0.212	0.723	pass

4.2. Within-region: the thermal increment replicates in five regions

Adding the six thermal predictors to the baseline raises spatially blocked
505 out-of-fold ROC-AUC in every region. The increment’s bootstrap interval
excludes zero in all five regions at 1 km and at 5 km blocking. Those are the
two scales this design supports as intervals. At 10 km the point estimates
hold, from +0.048 to +0.154. They rest on 6 to 33 positive-carrying blocks,
so they are reported as indicative rather than as intervals. The table note
510 gives the reason.

Table note (resampling units). The bootstrap resamples spatial blocks,
so what bounds an interval’s reliability is the number of blocks that carry
at least one burned cell. Those counts fall quickly as blocks coarsen. At
2 cells they are 235 (Manavgat), 302 (Bejís), 843 (Muğla), 716 (Evia) and
515 192 (Montiferru), out of 5 439, 3 967, 11 316, 2 566 and 743 blocks. At 10
cells they are 28, 19, 70, 41 and 16, out of 237, 176, 576, 155 and 35. At
20 cells they are **12, 6, 33, 15 and 6**, out of 60, 48, 167, 50 and 12. No
bootstrap replicate was invalid at any block size, with one exception. At
20 cells Bejís had 5 of 1000 replicates single-class. That is the symptom
520 of the six positive-carrying blocks just reported, and a further reason to
read the row as indicative. An equal-tailed percentile interval built on six
positive-carrying blocks has no meaningful coverage. Montiferru at 20 cells
also feeds only 12 groups into a 5-fold grouped split, so its models train on
about ten blocks each. **The 20-cell row of this table should be read**
525 **as indicative rather than as an interval.** The 10-cell row is the coarsest
blocking this design supports properly, because every region there has 16
to 70 positive-carrying blocks. The increment holds at that scale in all five

Table 2: Within-region baseline versus thermal performance and block-size robustness. Primary (TSG) population; spatially blocked 5-fold CV (Section 3.8); paired spatial-block bootstrap, 1000 replicates. Block sizes 2/10/20 cells $\approx$ 1/5/10 km.

Region	Block ($\approx$ scale)	Baseline AUC	Thermal AUC	Δ AUC	Δ AUC 95% CI
Manavgat 2021	2 ($\sim$ 1 km)	0.803	0.870	+0.067	[+0.055, +0.079]
	10 ($\sim$ 5 km)	0.748	0.797	+0.050	[+0.023, +0.077]
	20 ($\sim$ 10 km)	0.683	0.731	+0.048	[+0.014, +0.085]
Bejís 2022	2 ($\sim$ 1 km)	0.862	0.918	+0.056	[+0.048, +0.065]
	10 ($\sim$ 5 km)	0.779	0.825	+0.045	[+0.018, +0.069]
	20 ($\sim$ 10 km)	0.739	0.795	+0.057	[+0.031, +0.090]
Muğla 2021	2 ($\sim$ 1 km)	0.743	0.859	+0.116	[+0.106, +0.125]
	10 ($\sim$ 5 km)	0.698	0.777	+0.079	[+0.050, +0.105]
	20 ($\sim$ 10 km)	0.673	0.733	+0.061	[+0.030, +0.094]
North Evia 2021 (ext.)	2 ($\sim$ 1 km)	0.759	0.912	+0.153	[+0.142, +0.166]
	10 ($\sim$ 5 km)	0.716	0.864	+0.148	[+0.119, +0.183]
	20 ($\sim$ 10 km)	0.679	0.833	+0.154	[+0.124, +0.189]
Montiferu 2021	2 ($\sim$ 1 km)	0.781	0.883	+0.101	[+0.080, +0.125]
	10 ($\sim$ 5 km)	0.620	0.720	+0.100	[+0.017, +0.186]
	20 ($\sim$ 10 km)	0.555	0.681	+0.126	[+0.053, +0.228]

regions. Source: `paper/referee2_numbers.md`, block C, counted from the frozen per-cell prediction tables.

530 This is the first half of the trade-off. It is not itself novel, and it is reported because the second half is measured against it.

4.3. Cross-region transfer, and what label-free adaptation does to it

Raw transfer is heterogeneous and includes anti-predictive directions. Raw target AUC spans 0.326 to 0.686. Twelve of 20 directions are
535 above chance with CI support. Six are *below* chance with CI support: both directions of Manavgat to Bejís and of Manavgat to Muğla, plus Bejís to Evia and Evia to Bejís. Two intervals span 0.5. Even the best raw transfer, Evia to Manavgat at 0.686, stays far below that target’s own within-region thermal performance of 0.870. Across all directions the raw deficit against
540 the within-region reference is 0.184 to 0.592 AUC.

Those counts belong to the 2-cell blocking of Table 4. At the more conservative 10-cell blocking the same points give 9 above, 4 below and 7 uncertain, and no direction changes side of the chance line (Section 4.7). Four of the six below-chance directions keep their support there. Bejís to Manavgat and
545 Manavgat to Muğla lose it and carry no verdict. The qualitative statement is unchanged. The counts should not be read as exact.

The static baseline does not transfer either. This is the control for the reading the rest of the paper invites, and it constrains that reading sharply. The same twenty directions were run with the terrain, fuel and
550 greenness baseline alone. The mean target AUC is **0.537**, against **0.541** for the thermal model. Static attributes of a place are the class Dimarco et al. transfer successfully, and are the class this paper’s framing treats as

portable. Here that class is itself barely above chance. The paired per-direction contrast is given in Section 4.6b. What matters here is that the transfer failure below is not specific to the dynamic block. A baseline that
555 does not travel, plus pre-fire thermal state, gives a model that does not travel.

The thermal block’s paired contribution to transfer, with its interval. The two matrices were differenced direction by direction. The mean is **+0.004**. The directions are not independent, because each region
560 appears in eight of the twenty. The interval therefore depends on what is treated as the resampling unit. All four units the design permits give the same answer:

Table 3:

Resampling unit	n	95 % interval on the mean paired contribution
Directions, naive	20	[−0.027, +0.034]
Unordered pairs, cluster bootstrap	10	[−0.028, +0.036]
Unordered pairs, t on pair means	10	[−0.034, +0.042]
Regions, leave-one-out jackknife	5	[−0.037, +0.046]

Every interval spans zero, and the point estimate is a small fraction of each width. The per-region jackknife shows how little the mean is anchored.
565 Holding out Manavgat, Bejís, Muğla, Evia and Montiferru in turn gives +0.0148, +0.0021, +0.0065, **−0.0081** and +0.0060. **Dropping Evia alone reverses the sign of the headline.** None of these four units propagates within-direction sampling variability. They resample between directions only.

Label-blind adaptation compresses the matrix toward chance rather than repairing it. Under region-wise z-scoring the twenty directions
570

Table 4: Cross-region transfer matrix, thermal model, TSG population. Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after region-wise z-scoring ($\lambda = 10^{-5}$).

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.326 [0.305, 0.349]	0.477 [0.451, 0.502]	0.511 [0.484, 0.534]
Bejís→Manavgat	0.444 [0.408, 0.480]	0.457 [0.420, 0.497]	0.555 [0.528, 0.583]
Manavgat→Muğla	0.470 [0.451, 0.488]	0.431 [0.411, 0.449]	0.443 [0.423, 0.461]
Muğla→Manavgat	0.401 [0.378, 0.426]	0.559 [0.531, 0.587]	0.560 [0.535, 0.587]
Manavgat→Evia	0.613 [0.593, 0.631]	0.542 [0.520, 0.565]	0.539 [0.518, 0.561]
Evia→Manavgat	0.686 [0.653, 0.716]	0.516 [0.489, 0.544]	0.527 [0.500, 0.553]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.512, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.549 [0.524, 0.575]	0.549 [0.525, 0.573]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.567 [0.539, 0.594]	0.573 [0.538, 0.609]	0.606 [0.574, 0.639]
Manavgat→Montiferru	0.533 [0.488, 0.580]	0.586 [0.540, 0.629]	0.592 [0.550, 0.631]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.581 [0.539, 0.623]

span 0.431 to 0.630 and under CORAL 0.443 to 0.624, roughly half the raw spread, with no adapted direction exceeding 0.631 against within-region references of 0.859 to 0.918. Adaptation raises the failing directions and degrades most of those that already transferred. Taking the better of the two adaptations per direction, 14 of the 20 end closer to chance than they began
575 and 6 end further from it; five of those six involve Montiferru, the smallest and last-added region, and move upward, while the sixth is Manavgat→Muğla moving downward from 0.470 to 0.443. The 14 to 6 split should be read at the precision of limitation (x) in Section 5.9, since Bejís→Manavgat is
580 counted as compressed on a margin of 0.001.

In the six directions where raw transfer was below chance, the best label-free method recovers at most 34 % of the gap to the within-region reference, so the remaining conditional fraction is at least 0.66 everywhere. Seven directions show *negative* recovery, meaning adaptation moves the score away
585 from the reference; in six of those raw transfer was already above chance and adaptation destroyed that advantage. Label-free alignment therefore does not act as a repair mechanism.

4.4. *Transferability diagnostics: only conditional similarity orders transfer*

Twenty candidate diagnostics from four families were each rank-correlated
590 with the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap.

Only two diagnostics have intervals excluding zero, and both are conditional. The stronger is the sign-agreement fraction over interval-supported features, at $\rho = +0.84$ [+0.58, +0.88]. The cosine variant reaches
595 +0.81. No marginal measure was shown to order the matrix. That includes

Table 5: Transfer-gap decomposition (four-AOI set, 12 directions). Within = target’s within-region thermal AUC; best adapted = the better of z-score/CORAL; recovered fraction = (adapted – raw)/(within – raw), signed and unclipped, with paired bootstrap CI (1000 replicates). Montiferru directions are not part of this decomposition (per-pair absolute decompositions exist without fraction CIs). The status column asks whether the *adapted* value clears chance and uses the 2-cell adapted intervals of Table 4. The adapted arms were not recomputed at the coarser blocking of Appendix A(c), which covers the raw arm and the paired delta only.

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Manavgat→Bejís	0.918	0.326	0.511 (CORAL)	+0.31 [+0.28, +0.34]	recovery, chance not excluded
Bejís→Manavgat	0.870	0.444	0.555 (CORAL)	+0.26 [+0.18, +0.34]	recovery above chance
Muğla→Manavgat	0.870	0.401	0.560 (CORAL)	+0.34 [+0.28, +0.40]	recovery above chance
Bejís→Evia	0.912	0.383	0.532 (z-score)	+0.28 [+0.23, +0.32]	recovery above chance
Evia→Bejís	0.918	0.448	0.550 (z-score)	+0.22 [+0.16, +0.27]	recovery above chance
Manavgat→Muğla	0.859	0.470	0.443 (CORAL)	−0.07 [−0.12, −0.03]	negative recovery
Muğla→Bejís	0.918	0.583	0.560 (CORAL)	−0.07 [−0.16, +0.01]	negative recovery
Manavgat→Evia	0.912	0.613	0.542 (z-score)	−0.23 [−0.32, −0.14]	negative recovery
Evia→Manavgat	0.870	0.686	0.527 (CORAL)	−0.86 [−1.18, −0.60]	negative recovery
Evia→Muğla	0.859	0.577	0.530 (CORAL)	−0.17 [−0.22, −0.11]	negative recovery
Muğla→Evia	0.912	0.653	0.563 (CORAL)	−0.35 [−0.43, −0.27]	negative recovery
Bejís→Muğla	0.859	0.618	0.518 (z-score)	−0.42 [−0.51, −0.34]	negative recovery

Table 6: Transferability diagnostics versus raw thermal transfer, by family. Spearman ρ against raw transfer AUC with pair-based bootstrap 95 % CIs. The strongest member of each family in its expected direction is named. All twenty individual diagnostics are in Appendix B, Table B1.

Family	Diagnostics	Strongest member, in the expected direction	Any CI excluding 0
P(y x) conditional	6	+0.84 [+0.58, +0.88] agreement fraction, supported features	yes, 2 of 6
P(x y=1) niche overlap	5	+0.24 [−0.45, +0.74] Schoener’s D, 1-D mean	no
P(x) marginal	6	−0.32 [−0.78, +0.33] domain-classifier AUC	no
P(y) regime structure	2	+0.29 [−0.38, +0.74] log effective-N distance	no
geographic	1	−0.24 [−0.84, +0.73] centroid geodesic distance	no

area-of-applicability-style dissimilarity in predictor space, climatic distance and geographic distance. The niche-overlap and regime families were not shown to order it either. The learned domain classifier is at ceiling, separating source from target at $AUC \geq 0.96$ for every pair. It always succeeds, which
600 is why it carries no ordering information.

Two limits are stated with the result rather than after it. The first is size. The index’s tie structure caps the achievable Spearman at +0.861, so the observed +0.840 sits essentially on that ceiling. Its exact one-sided permutation p is 0.0060, the smallest this tie structure can produce, against
605 a Bonferroni threshold of 0.0026 over the nineteen computed variants. No outcome of this diagnostic could have cleared family-wise correction on ten effective pairs. The second limit is labels. Signed associations need burned labels in both regions. The family that works is therefore not available before deployment, while the family that fails is.

610 A third limit is about selection, and it is the sharpest of the three. The two rows that clear zero are the *supported-feature* variants. Their feature subset is chosen by whether two regions’ bootstrap intervals happen to be disjoint. That is a data-dependent selection, made on the same data, with no correction. The unselected counterparts over all nine features are $\rho =$
615 $+0.50 [-0.17, +0.83]$ for the cosine and $\rho = +0.18 [-0.40, +0.72]$ for the agreement count. Both span zero. The result lives in the selection step, and is reported as such.

The families are compared on unequal samples, and equalising them does not change the ordering. The marginal, applicability,
620 climatic and geographic rows sit on twelve directions, because those diag-

nostics exist only for the four-region subset. The supported-conditional rows sit on sixteen and the rest on twenty. A reader may reasonably ask whether the marginal family’s failure is a statement about power rather than about diagnostics. Every row was therefore recomputed on the common twelve di-
625 rections. The published values reproduce to 4.8×10^{-5} . The conditional rows still lead, at $+0.87 [+0.65, +0.88]$ and $+0.85 [+0.43, +0.88]$, and every marginal row still spans zero. The ordering is not an artefact of unequal samples.

4.5. *The contrast pair: similarity is not sufficient*

630 The clearest single view of Table 6 needs no ranking at all. Manavgat and Muğla lie in the same country and the same fire year. They are 307 km apart by the centroid geodesic distance this paper uses as a diagnostic, and their nearest boundaries are 191 km apart. Their burned cells occupy the most similar environmental envelope of any pair in the matrix, with
635 per-feature Schoener’s D of 0.77 to 0.89. Yet five of nine feature-response directions point opposite ways. Elevation is one of them, at a signed AUC of 0.374 [0.289, 0.471] in Manavgat against 0.611 [0.532, 0.690] in Muğla, with disjoint intervals. All four absolute thermal channels are among the others. Transfer is below chance in both directions at the point estimate, at 0.470
640 and 0.401. Both directions also sit inside the nominal area of applicability, at 0.875 and 0.531 of target cells.

Bejís and Montiferru sit at the opposite extreme. Their burned envelopes barely overlap, and they carry the most dissimilar values on every overlap measure. They transfer above chance in both directions. That half of the
645 contrast rests on point estimates: neither direction carries a verdict at 5

km blocking, and the marginal applicability audit was never produced for Montiferru. The claim is one of *sufficiency*. Similarity does not guarantee transfer, and dissimilarity does not preclude it. The claim rests on two coexisting counterexamples, so it does not depend on the number of pairs
650 available.

4.6. Interventions: pooling and feature removal

(a) **Pooled multi-region training.** Training on the pooled primary populations of the other four regions never beats the best single-source transfer for any target, and stays 0.28 to 0.50 AUC below the within-region ceiling;
655 for two targets it falls below the pairwise mean and below chance, though only Bejís is below chance with interval support, at 0.417 [0.369, 0.467]. Aggregation does not manufacture the missing conditional information.

(b) **Removing the direction-reversing features.** The two predictors whose signed association reverses between regions are **elevation_mean** and
660 **lst_anomaly_mean**. Retraining without them costs **-0.081** of mean within-region AUC, supported in every region (per-region deltas -0.060 , -0.130 , -0.074 , -0.063 , -0.079 , every interval entirely below zero), and changes mean transfer by **+0.014** [**-0.017**, **+0.045**], which spans zero.

Two qualifications belong with those numbers rather than after them.
665 First, the debit is not the thermal block’s. Dropping elevation alone accounts for -0.061 of it, from 0.888 to 0.827. Dropping the LST anomaly alone accounts for -0.013 , from 0.888 to 0.875. Roughly three quarters of the cost is therefore the removal of a **baseline** terrain variable, not of a thermal channel. Second, both figures are post-selection. The two predictors were
670 chosen because they reverse, using the same data on which the -0.081 and

the +0.014 are then estimated. No correction for that selection is applied.

Table 7:

Configuration	Mean within-region AUC	Mean transfer AUC
full	0.888	0.541
drop <code>elevation_mean</code>	0.827	0.546
drop <code>1st_anomaly_mean</code>	0.875	0.544
drop both	0.807	0.556

What this measures is a local cost with no compensating transfer gain. It does not measure an exchange, because the transfer side is a null on both arms. The thermal block’s own contribution to transfer is +0.004, with
675 an interval spanning zero (Section 4.3). Removing the reversing predictors returns +0.014, with an interval spanning zero. Two nulls on the portability axis are not a price paid.

4.7. Sensitivity analyses

Five design choices were varied with everything else held fixed. They
680 are the Evia AOI and its prevalence, the CORAL regularisation constant, the blocking scale, the closure date of the predictor window, and the quality screening of the coarse thermal input. None changes a conclusion above. Two bound how the results should be read, so they are carried into the main text here.

685 Coarsening the blocks from 1 km to 5 km moves the verdict counts from ten positive, seven negative and three uncertain to six, four and ten. Support is removed from six verdicts and added to none. The point estimates are unchanged, but that is an identity rather than a result. The blocking scale

is the resampling unit, and it cannot move an estimate computed once over
690 all target cells.

Manavgat’s whole downstream chain was then rebuilt from a quality-screened MODIS input. That changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C. No signed univariate association moves by more than +0.0003. This closes the one processing-artefact candidate for
695 that region’s behaviour. Appendix A reports all five arms.

4.8. The same geography, a second fire: reversal with place held constant

Every result above compares different places. Muğla admits a stricter test, because a second fire burned inside the identical AOI on the identical analysis grid. The 2021 event has a predictor window closing on 28 July. The 2022
700 event has a matched window closing on 20 June. Region, bounding box, cell definition, feature registry and processing chain are the same in both arms. The design holds place fixed. It does not hold season fixed, because the 2022 fire ignites about six weeks earlier, and it does not hold the population fixed, because the 2022 arm is defined by removing the 2021 scar.

Elevation reverses with bootstrap support. In 2021 higher ground
705 burned preferentially, at 0.611 with its interval entirely above 0.5; in 2022 lower ground did, at 0.296 with its interval entirely below. The two intervals are disjoint, 0.532 against 0.355, and the difference is -0.317 $[-0.414, -0.220]$. The same predictor, the same region, the same grid, and an association that points the opposite way in two fires eleven months apart. The
710 four absolute thermal channels change side as well, from bootstrap-supported lower-values-burn in 2021 to higher-values-burn point estimates in 2022.

Two structural properties must be read alongside these numbers, because

Table 8: Signed univariate feature-burned AUC, Muğla 2021 versus 2022. Raw AUC against **burned**, never folded to $\max(\text{AUC}, 1 - \text{AUC})$; 10-cell (≈ 5 km) spatial-block bootstrap, 1,000 replicates, seed 42 (Section 3.14). Analysis population 41,730 rows / 2,911 burned (2021) and 38,790 rows / 331 burned (2022). **Positive-carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm.** Table 2’s note sets sixteen as the floor this design supports at that blocking, so the 2022 intervals here fall below the paper’s own standard and are read as indicative, exactly as the 20-cell row of Table 2 is. The 2022 arm is additionally a single compact scar, so its eleven blocks are contiguous. Read from `paper/step9g_raw/.../mugla_2021__mugla_2022_event_relative/step9g_direction_reversal_table.csv`.

Feature	2021 AUC [95 % CI]	2022 AUC [95 % CI]	Reversal
elevation_mean	0.611 [0.532, 0.690]	0.296 [0.230, 0.355]	bootstrap- supported
current_lst_mean	0.325 [0.271, 0.382]	0.515 [0.434, 0.580]	point only
current_tvdi_mean	0.336 [0.275, 0.398]	0.594 [0.475, 0.674]	point only
downscaled_lst_mean	0.307 [0.253, 0.366]	0.508 [0.435, 0.571]	point only
fused_lst_mean	0.325 [0.272, 0.383]	0.519 [0.436, 0.583]	point only
ndvi_mean	0.662 [0.616, 0.704]	0.707 [0.624, 0.777]	none
slope_mean	0.637 [0.582, 0.686]	0.558 [0.468, 0.634]	none
lst_anomaly_mean	0.485 [0.395, 0.566]	0.380 [0.249, 0.502]	none
tvdi_difference_mean	0.490 [0.396, 0.575]	0.397 [0.265, 0.514]	none

they have no analogue in the twenty-direction matrix. The two arms are
715 not disjoint samples: they share 38,789 of the 2022 arm’s 38,790 cells, and
elevation, slope and land cover are identical to the digit across all 73,098
grid cells, so only NDVI and the six thermal channels carry new information
between them. And the removal is the target’s own positive class, so in the
2022 to 2021 direction not one target positive is present in the source training
720 population while 38,789 of 38,819 target negatives are, and membership of
the source training set alone separates the 2021 target’s classes at ROC-
AUC 0.9996. We therefore cannot bound what that asymmetry does to the
two transfer numbers, only state that a direction with this structure is not
comparable to the twenty between-region directions. The elevation reversal is
725 the finding; the structural asymmetry is a competing explanation this design
cannot exclude. The known part of the bias runs the safe way: the removed
cells are high, with a median of 563 m, and unburned in 2022, so removing
them raises the 2022 elevation AUC and makes the reversal smaller rather
than larger.

730 4.9. *What target labels cost: the recovery curve*

Everything above measures a failure; this prices it. If the residual gap
is conditional and label-free alignment cannot close it, the missing resource
is target-conditional information, and the direct way to supply it is target
labels. A frozen few-shot diagnostic answers how many, for three regions
735 across all six ordered directions, using one 10-cell (~ 5 km) spatial block as
the unit of labelling effort and reading recovery against a matched target-only
ceiling of 0.777 to 0.824.

Thirty-two labelled blocks recover **85 to 89 % of that ceiling in three**

of the six directions, two of which started below chance; 51 to 57 % in two
740 more, the directions where the conditional gap is widest; and 30 % in the
sixth. That budget is 7 to 20 % of the target’s natural-vegetation population,
roughly 800 km, so it is a real answer and not a cheap one. Two properties
of the measurement bound it and are stated in the supplement rather than
buried: at the top budget the selection pool is nearly exhausted, so the
745 narrow upper-budget intervals reflect saturation rather than precision, and
the labelled blocks are drawn from the event being predicted, which is not a
resource available before that event burns. The protocol, the full curve and
the limits are in Supplement S1.

5. Discussion

750 5.1. Principal findings

Three results carry this paper. First, the pre-fire thermal block raises
spatially blocked within-region ROC-AUC by +0.056 to +0.153 in every one
of five Mediterranean regions, with bootstrap support at 1 km and 5 km
blocking, and contributes +0.004 $[-0.028, +0.036]$ across twenty ordered
755 transfer directions, indistinguishable from zero and with a sign that varies
by pair. The static baseline transfers no better, at a mean of 0.537 against
0.541, so the failure is a property of this matrix rather than of the dynamic
block specifically. Removing the two reversing predictors, elevation and the
LST anomaly, costs -0.081 of within-region skill, supported in every region
760 and three quarters attributable to elevation, and changes transfer by +0.014
 $[-0.017, +0.045]$, which also spans zero. Second, of twenty candidate trans-
ferability diagnostics only two have intervals excluding zero and both are

conditional, while no marginal measure was shown to order the matrix, including the area-of-applicability family that is the only one runnable before
765 deployment. Third, the mechanism is a reversal in the sign of the predictor-burning association, and it persists inside one study area across two fires.

5.2. *Why the thermal increment is real but local*

The within-region increment is not an artefact to be explained away. It replicates in five independent regions, survives coarsening of the spatial
770 blocks to ~ 5 km with its interval intact, persists in the secondary all-valid population, and survives a predictor window closed 7 and 14 days earlier in all five regions. The direction of that last result is region-specific and should not be overstated: the contribution strengthens in Bejís ($0.058 \rightarrow 0.079$ at 14 days) and Muğla ($0.115 \rightarrow 0.128$), is flat in Montiferru, and **weakens**
775 **monotonically in Evia** ($0.156 \rightarrow 0.149 \rightarrow 0.135$). What holds everywhere is survival, not improvement. The transfer failure is therefore not evidence that the thermal signal is spurious. It is evidence that the fitted relationship is *local*.

The natural objection is that different Mediterranean regions are simply
780 different systems, so that a predictor meaning one thing in one place and another elsewhere is not instability but a comparison of two different systems. The two Muğla events answer it directly: region, AOI, grid, feature registry and processing chain identical, static predictors identical cell by cell, and elevation’s association with burning reversing from 0.611 [$0.532, 0.690$] to
785 0.296 [$0.230, 0.355$], a difference of -0.317 [$-0.414, -0.220$] (Section 4.8). Holding geography fixed does not stabilise the direction of the relationship.

Three caveats attach and none is dismissed. Season and year are con-

founded, and **that confound cannot be resolved in this study area**:
the events sit 42 days apart in median burn day-of-year, neither year con-
790 tains a second event at the other’s phase, and a calendar-matched arm would
carry nine burned cells against a gate minimum of thirty. The population is
not held fixed either, the 2022 arm being the 2021 arm with the 2021 scar
removed. And the 2022 arm rests on one compact scar and eleven positive-
carrying 5 km blocks. What survives all three is the narrower reading that
795 *something* reverses the elevation-burning association with geography, grid
and pipeline held fixed, which is why Section 4.8 is reported as corroboration
rather than as the load-bearing evidence for the mechanism.

The sharpest supported reversal belongs to elevation, a static predictor,
so instability is a property of the predictor-to-burning mapping generally
800 rather than of thermal channels specifically. This is the same conclusion the
baseline transfer arm reaches from the other direction: the instability is not
the thermal block’s peculiarity.

5.3. *Why the similarity-based diagnostics fail*

A diagnostic built on distance in predictor space asks one question: do
805 the target’s predictor values look like the training data’s? That is not the
question that matters when the failure is conditional. A target region can
sit well inside the training envelope while the relationship between those
predictors and burning points the other way. Manavgat to Muğla is exactly
that case, with 0.875 of target cells inside the weighted area of applicability
810 and transfer below chance.

The same holds for the niche-overlap and regime families. At the point
estimates, the pair with the highest burned-niche overlap in the matrix

fails in both directions while the pair with the lowest transfers in both, so no monotone function of overlap can order the outcomes. The qualifier is
815 not decorative: at the 5 km blocking this paper otherwise defends, neither Bejís↔Montiferru direction carries a verdict and Manavgat→Muğla loses its below-chance support, so this contrast is a statement about point estimates and is made as one. The domain classifier is at ceiling for every pair, which is informative in itself: the regions are trivially separable in predictor space
820 whether or not their transfer works, so separability carries no ordering information here.

5.4. *What the conditional diagnostic is, and what it is not*

The sign-agreement index reaches $\rho = +0.84$ [+0.58, +0.88] against observed transfer. Three limits are reported with it rather than after it. Its
825 tie structure caps the achievable Spearman at +0.861, so the observed value sits essentially on its own ceiling and the ranking it supports is coarse. Its exact one-sided permutation p is 0.0060, the smallest that tie structure can produce, against a Bonferroni threshold of 0.0026 over nineteen computed variants. No outcome could have cleared family-wise correction on ten ef-
830 fective pairs. And it needs burned labels in both regions, which makes it a mechanism diagnosis rather than a pre-deployment screen.

That last limit is the substantive one, and it is why the paper’s practical conclusion is not "use this index instead". It is that the family of diagnostics that can be run before deployment is the family that fails here, and the
835 information that would order transfer is the information a practitioner does not have when the decision is made. The honest statement is that these twenty candidates were **not shown to order transfer**, not that they are

shown incapable of it.

5.5. *What the two interventions do and do not show*

840 Both interventions show the same shape. Pooling four regions never beats the best single-source transfer for any target and stays 0.28 to 0.50 below the within-region ceiling, so aggregation does not manufacture the missing conditional information. Removing the reversing predictors costs -0.081 of within-region skill with interval support in every region, and changes mean
845 transfer by $+0.014$ $[-0.017, +0.045]$, which is not distinguishable from zero.

That pair of numbers is easy to read as an exchange, and it is not one. The thermal block's own contribution to transfer is $+0.004$, with an interval spanning zero. Removing the reversing predictors returns $+0.014$, with an interval spanning zero. Both arms are null on the portability axis. What
850 the interventions measure is a local cost and no compensating transfer gain. That is not a conservation law, and it is not a rate at which local skill can be sold for portability. No such rate is estimated here.

5.6. *The regime hypothesis, reported as it happened*

A regime-structure explanation was stated in advance and the data con-
855 firmed the null. The regime-distance correlation has the wrong sign at the point estimate, the most regime-similar pair fails in both directions, and the most regime-different pair transfers above chance. One mundane explanation can be set aside: cutting Muğla, by far the largest population, to Manavgat's cell count and separately to its positive count leaves Muğla's transfer
860 behaviour inside the subsampling range in both roles. The error was in the

hypothesised grouping, not in the data, and with ten pairs this cannot refute regime typology [2] in general.

5.7. *The empirical contrast with Dimarco et al.*

Dimarco et al. [7] transfer successfully across a comparable Mediterranean
865 design and we do not, and the two results are not in conflict. Their predictors are attributes of a place, ours describe the state of a surface in one season, which suggests that the relationship between domain similarity and transfer success is predictor-class dependent.

That reading is one of at least two, and we cannot separate them here.
870 Their response variable is human-driven **ignition**; ours is burned **area**. Ignition likelihood is dominated by human access and activity, stable and near-universal drivers, whereas burned extent is dominated by spread. A predictor-class explanation and a response-variable explanation are therefore confounded in this comparison. Our own data speak against a simple
875 predictor-class reading in any case: the static baseline transfers at a mean of 0.537 here, so within this cohort the place-attribute class does not travel either.

5.8. *Implications*

For practice, transfer skill has to be *measured* rather than inferred from
880 similarity, and the pre-deployment diagnostics currently used for that inference did not order it here. Where a model must be moved, the resource that closes the gap is target labels: thirty-two labelled 5 km blocks recover 85 to 89 % of the target’s own ceiling in three of six directions, and 30 to 57 % in the rest. That is a real answer and not a cheap one, being 7 to 20 % of

885 the target’s natural-vegetation population, and at small budgets the same intervention damages the direction that already transfers best.

For method development, the results point away from better unsupervised alignment. Two label-free methods applied carefully moved fourteen of twenty directions towards chance, and every direction they improved involves 890 the smallest region. A sign reversal is not a distribution mismatch, and no realignment of inputs can repair it.

5.9. Limitations

The transfer failure is a finding, not a limitation. The limitations are the boundaries on how far it generalises.

895 (i) **No meteorological covariates** enter the models, so we cannot say how the trade-off behaves for a mixed thermal-plus-weather predictor set.

(ii) **Temporal transfer is measured for one region only**, and even there year and seasonal phase are confounded, a confound that cannot be resolved in this study area for the reason given in Section 5.2. Its 331 burned 900 cells also leave the thermal reversals unresolved at interval level, and the pair holds place fixed but not population. Those two arms are additionally the only transfer directions here computed by us rather than read from the pipeline author’s frozen export, with his unmodified code and the same pinned environment.

905 (iii) **All labels derive from a single burned-area product**, MCD64A1 [9], whose omission and commission characteristics [3] bound every model evaluated here. No second burned-area product covers 2021 and 2022 at this resolution; the companion paper reports what an independent active-fire observation says about the omission concern.

910 (iv) **Evia remains the most imbalance-atypical population** even after the AOI extension, at a TSG prevalence of 0.287 against 0.038 to 0.072 in three of the others.

(v) **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology, and distinguishing them requires
915 multi-year labels.

(vi) **Cross-region point estimates carry an implementation tolerance** of roughly ± 0.02 to 0.03 across scikit-learn versions. All reported numbers are fixed to one verified version, but exact reproduction elsewhere requires the archived environment.

920 (vii) **The diagnostic correlations rest on an effective sample of ten region pairs.** Both the successes and the failures of Section 4.4 should be read at that power.

(viii) **Manavgat’s atypical transfer behaviour remains unexplained.** It is where the conditional diagnosis bites hardest and where feature removal
925 recovers most. Two candidates have now been tested and neither survives. Its predictor window was not meteorologically extreme. The second candidate was the quality-screening difference in its coarse thermal input. It was the more worrying of the two, because the change it induces correlates with elevation at +0.615. It does not propagate. The region’s entire downstream
930 chain was rebuilt from a quality-screened MODIS input. That changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C. No signed univariate association moves by more than +0.0003. Elevation’s stays at 0.374 [0.290, 0.472] in both arms, and the population is unchanged. The result is structural rather than fortunate. Elevation is a DEM variable the screening

935 cannot touch, and fusion falls back on the MODIS-derived surface across
only 2.14 points of coverage. With one fire season per region the remaining
candidates are not separable in this design.

(ix) **The interval-support counts are less stable than the point estimates behind them.** Several verdicts sit within a thousandth of their
940 reference value, and at 1 km blocking the published split of ten positive,
seven negative and three uncertain turns on a lower bound of -0.00045 . The
point estimates and the sign pattern are stable; the counts are not. Every
sentence in this paper that leans on an exact count of supported directions
should be read at that precision.

945 6. Conclusions

Pre-fire thermal dryness adds a real and repeatable increment to burned-
area discrimination. Across five Mediterranean regions, ROC-AUC was raised
by $+0.06$ to $+0.15$ over a static and near-static terrain, fuel and greenness
baseline. That baseline's only time-varying member is the vegetation-index
950 composite. The gain survived spatial blocking at about 5 km, the coarsest
scale this design supports as an interval. It also survived a predictor window
closed up to two weeks before the first labelled burning.

That skill is local. Paired per direction, the same predictor block adds
nothing distinguishable from zero to cross-region transfer. The mean is
955 $+0.004$ over twenty directions and its interval spans zero under all four
between-direction resampling units we computed. Its contribution changes
sign from one direction to another. The paired deltas run from -0.148 to
 $+0.132$, twelve positive and eight negative. At the conservative 5 km block-

ing, five to six directions are helped and three to four are harmed with interval
960 support, with the rest carrying no verdict. The block is also the swing factor
at the chance line.

The control that fixes the meaning of all this is the baseline arm, which
transfers at a mean of 0.537 against the thermal model’s 0.541. The static
predictor class is not portable here either, so the honest summary is that
965 nothing in this feature space travels, not that dynamic state is uniquely frag-
ile. Removing the two reversing predictors, elevation and the LST anomaly,
costs -0.081 of mean within-region skill, with interval support in every re-
gion and roughly three quarters of it attributable to elevation, a baseline
terrain variable; it changes mean transfer by $+0.014$, an estimate whose in-
970 terval spans zero. A local cost is measured. No compensating transfer gain
is, on either arm, so no exchange between the two is demonstrated.

The failure is conditional. The direction of the link between dryness
and burning changes from one region to another. None of the similarity
diagnostics tested here ordered the transfer matrix: predictor-space distance,
975 domain separability, niche overlap and regime structure all failed. On ten
effective pairs those null results mean not shown to order transfer, rather
than shown not to. It is also why label-blind adaptation compresses transfer
towards chance instead of repairing it, and why pooled multi-region training
does not escape it.

980 The practical implication is a change in what is checked before a dynamic-
state fire model is transferred. The question is not whether the target region
lies inside the source’s environmental envelope. In our matrix, the pair with
the highest envelope overlap failed in both directions, and the pair with the

lowest overlap transferred in both. Both statements are made at the point
985 estimate. The question is whether the signed feature-response directions
agree.

That check requires a labelled probe in the target region, and so, on
inspection, do the niche and regime diagnostics. Only the marginal family
can be computed before any target label exists, and it is the family that
990 failed. How large the probe has to be is not established here, so no label
budget is claimed. A signed association with a usable interval may need
fewer labels than a refitted model, but that was not tested. What is clear is
that the supervised recalibration that label-free alignment cannot deliver is
not cheap. Section 4.9 prices it for three regions: thirty-two labelled 5 km
995 blocks recovered 85 to 89 % of the target ceiling in three of six directions, 51
to 57 % in two more and 30 % in the sixth, and at its top budget the labelled
set already holds most of one target region’s burned cells. Where no target
labels exist, the transfer performance of such models should be treated as
unknown rather than inferred from similarity.

1000 For model builders, local skill and portability should be reported sepa-
rately rather than assumed to travel together. A predictor block worth a
large within-region increment may contribute nothing distinguishable from
zero across regions, with a sign that varies by pair, and reporting only within-
region validation hides that completely. Three extensions are left for fu-
1005 ture work: temporal transfer, meteorological covariates, and physically nor-
malised dryness variables. Each is a step towards the self-calibrating thermal
monitoring system that motivated this study.

Appendix A. Appendix A. Sensitivity analyses

Each arm below varies one design choice and leaves everything else fixed.
1010 None changes a conclusion in the main text. They are reported so that a reader can see which choices were tested, and what each was worth. Arm (f) is the exception in one respect: it is not a robustness check but a test of a claim the introduction makes, and the claim is not upheld.

(a) **Evia AOI and prevalence.** The raw transfer arms were repeated
1015 with the legacy, high-prevalence Evia box. Every qualitative conclusion is unchanged. Thermal raw transfer AUCs move by up to 0.07, the largest being Evia to Bejís at 0.378 to 0.448, and no direction changes side of the chance line.

(b) **CORAL regularisation.** Over nine λ values from 0 to 10^{-1} on
1020 four directions, CORAL transfer AUC moves by at most 0.014 within any direction and 0.008 within the thermal family, so no CORAL-dependent conclusion here is sensitive to λ in that range. The companion paper reports the canonical $\lambda = 1$, which the released sweep omits.

(c) **Blocking scale.** Recomputing the transfer quantities at 10-cell ($\approx$
1025 5 km) blocking from the frozen per-cell predictions widens the intervals and moves the verdict counts, from ten positive, seven negative and three uncertain at 1 km to six, four and ten at 5 km. Coarser blocking therefore removes support from six verdicts and adds none.

The point estimates are unchanged. That is an identity rather than a
1030 result, and it should not be offered as robustness. The blocking scale is the bootstrap *resampling unit*, and each point estimate is computed once over all target cells, so no choice of block size could have moved one. The

comparison does establish two things. Every verdict that changes, changes towards "no verdict". And no direction crosses the chance line under the
1035 widened intervals. The counts are the fragile part of this paper. The sign pattern is not thereby shown to be robust. It is simply not tested by this variation.

(d) Predictor-window closure. The predictor window was closed 7 and 14 days earlier in all five regions. The thermal contribution stays positive
1040 and bootstrap-supported everywhere. The direction of the change is region-specific. It strengthens in Bejís, from 0.058 to 0.079 at 14 days, and in Muğla, from 0.115 to 0.128. It is flat in Montiferru. It weakens monotonically in Evia, from 0.156 to 0.149 to 0.135. What holds in every region is survival, not improvement. That is the claim made in Section 5.2.

(e) Quality screening of the coarse thermal input. Two of the
1045 five regions' MODIS inputs are quality-screened and three are not. The split follows export date rather than design, and it induces an elevation-correlated change at the input, at $r = +0.615$ in Manavgat. Manavgat's entire downstream chain was rebuilt from a quality-screened input. That
1050 changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C. No signed univariate association moves by more than +0.0003. Elevation stays at 0.374 [0.290, 0.472] in both arms, the population is unchanged, and the within-region increment moves from [+0.055, +0.079] to [+0.054, +0.077]. The reason is structural. Elevation is a DEM variable the screening
1055 cannot touch, and fusion falls back on the MODIS-derived surface across only 2.14 points of coverage. Details are in `paper/modis_qc_downstream_propagation.md`.

(f) **Normalised against absolute dryness channels.** Section 1.2 argues that an internally normalised index should be less exposed to absolute-
1060 temperature offsets between regions than raw land surface temperature. The thermal block contains both kinds, which allows the argument to be tested directly as a feature-set contrast. It is also the natural place to look for a constructive result, because the two channels that do **not** reverse sign between regions are exactly the normalised ones, `lst_anomaly_mean` and
1065 `tvdi_difference_mean`, while the four absolute channels are the ones that do.

Three feature sets were run over all twenty directions and all five within-region folds, with the classifier, population, folds and bootstrap held fixed. The harness aborts unless its reference configuration lands on the frozen ex-
1070 ports. It did exactly: the maximum absolute difference between the reference arm and the frozen transfer AUCs is 0.000000 across all twenty directions.

The advantage is not found. Per direction, the normalised set minus the absolute set has a mean of -0.0037 . It is positive in 12 of 20 directions and runs from -0.099 to $+0.094$, so the spread is an order of magnitude larger
1075 than the difference. The normalised channels transfer very slightly worse at the mean, and nothing here supports a design rule favouring anomaly-referenced dryness for portability.

Within region the two sub-blocks are indistinguishable, at 0.8551 and 0.8564, and each recovers most of the gap between the baseline’s 0.7896 and
1080 the full model’s 0.8883. What separates them in sign stability does not become transferable skill.

This arm strengthens the negative finding rather than softening it. Four

Table A.9:			
Feature set	Mean transfer AUC	Directions > 0.5	Mean within-region AUC
baseline only	0.5371	16	0.7896
baseline + normalised anomalies	0.5442	15	0.8551
baseline + absolute surface state	0.5479	13	0.8564
all ten features (reference)	0.5414	14	0.8883

differently constituted feature sets were tried. None clears a mean of 0.548 across regions, and all differ sharply within them. The baseline arm also gives an independent confirmation of the control reported in Section 4.3: its mean transfer of 0.5371 was recomputed here from the modelling datasets and matches the 0.537 read from the frozen per-direction export. Details are in `paper/anomaly_only_transfer.md`.

The comparison is between two sub-blocks of one thermal set on one cohort. It does not test normalised dryness indices in general, and it does not test a normalisation fitted against a pooled multi-region reference rather than each region’s own baseline years.

Appendix B. Appendix B. All twenty transferability diagnostics

Section 4.4 summarises this table by family. The full ranking is given
1095 here because the negative result is the point: a reader should be able to see
every candidate that was tried, not only the families' best members, and to
check that no diagnostic was dropped after it failed.

**Table B1. All transferability diagnostics versus raw thermal
transfer (20 ordered directions).** Spearman ρ with pair-based bootstrap
1100 95 % CIs. Exp. = expected sign. Rows with $n = 12$ exist only for the four-
AOI subset, because those diagnostics were never produced for Montiferru.
The supported-features conditional rows use the 16 directions (8 pairs) with
at least one CI-supported feature.

Table note. A null row means the diagnostic was **not shown to order**
1105 **transfer** on this design, not that it was shown incapable of ordering it.
With ten effective region pairs the power is low, and the intervals are wide
enough to admit moderate true correlations in either direction. The last row
is retained rather than deleted because it was computed: on two directions
the statistic has no meaningful value, and reporting that is more honest than
1110 dropping the variant.

Reading the two rows that clear zero. Both are supported-feature
variants, where the predictor subset is chosen by whether two regions' boot-
strap intervals happen to be disjoint. That is a data-dependent selection
made on the same data, uncorrected. Their all-nine-feature counterparts are
1115 in the table and both span zero. The result lives in the selection step, and
Section 4.4 says so.

Equal-sample check. The families sit on unequal samples: marginal,

Table B.10:

Diagnostic	Family	Exp.	n dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	$P(y x)$ conditional	+	16	+0.84 [+0.58, +0.88]	yes
Cosine, supported features	$P(y x)$ conditional	+	16	+0.81 [+0.33, +0.88]	yes
Cosine, all 9 features	$P(y x)$ conditional	+	20	+0.50 [−0.17, +0.83]	no
Vector Spearman, all 9	$P(y x)$ conditional	+	20	+0.27 [−0.36, +0.77]	no
Agreement count, all 9	$P(y x)$ conditional	+	20	+0.18 [−0.40, +0.72]	no
Schoener’s D, 1-D mean	$P(x y=1)$ niche	+	20	+0.24 [−0.45, +0.74]	no
Warren’s I, 1-D mean	$P(x y=1)$ niche	+	20	+0.22 [−0.42, +0.73]	no
Schoener’s D, PCA-2D	$P(x y=1)$ niche	+	20	+0.10 [−0.51, +0.68]	no
Warren’s I, PCA-2D	$P(x y=1)$ niche	+	20	−0.07 [−0.66, +0.49]	no
Mahalanobis, burned centroids	$P(x y=1)$ niche	−	20	−0.23 [−0.75, +0.44]	no
Domain-classifier AUC	$P(x)$ marginal	−	20	−0.32 [−0.78, +0.33]	no
Predictor-space mean dissimilarity	$P(x)$ marginal	−	12	−0.10 [−0.54, +0.43]	no
Predictor-space p95 dissimilarity	$P(x)$ marginal	−	12	−0.08 [−0.59, +0.49]	no
Fraction inside weighted AoA	$P(x)$ marginal	+	12	+0.22 [−0.48, +0.59]	no
Fraction inside	$P(x)$ marginal	+	12	+0.08 [−0.89, +0.43]	no

applicability, climatic and geographic rows on twelve directions, the supported-
conditional rows on sixteen, the rest on twenty. Recomputing every row on
1120 the common twelve directions reproduces the published values to 4.8×10^{-5}
and leaves the ordering unchanged — the conditional rows still lead at +0.87
[+0.65, +0.88] and +0.85 [+0.43, +0.88], every marginal row still spans zero.
Source: `paper/diagnostics_common_subset.md`.

Appendix C. Appendix C. Protocol detail

1125 These protocols are given here in full rather than in Methods, because
each is a specification a reader needs only when checking the corresponding
result, and neither is needed to follow the argument. Sections 3.2, 3.3, 3.11
and 3.14 state in summary what each does.

Appendix C.1. C.1 Burned-area label, the reconstructed analysis grid, and
1130 *the admissibility gate*

Labels come from MODIS MCD64A1 Collection 6.1 (`MODIS/061/MCD64A1`)
retrieved through Google Earth Engine [10], whose omission and commission
characteristics [3] bound every model here. The product is exported onto
the 30 m EPSG:4326 reference grid used by the rest of the pipeline, which
1135 duplicates each native ~ 500 m observation across the 30 m pixels beneath
it.

The analysis grid is therefore **reconstructed rather than native**: the
30 m grid is partitioned into non-overlapping blocks of `round(500 / 30) =`
`17 × 17` pixels, giving a nominal cell edge of 510 m, and each cell is identified
1140 by its integer block indices `row_500m` and `col_500m`. This approximates the
MODIS sinusoidal cell rather than reproducing it, being anchored to the

Landsat/EPSG:4326 reference grid, and marginal blocks are truncated. The cell is square in degrees but not on the ground, about 510 m north to south and 390 to 407 m east to west, so its area is 0.199 to 0.208 km² against a
1145 MODIS cell's 0.250 km²; every block-size label in this paper is therefore the north-south dimension. The companion paper develops what follows from that.

A cell's representative burn date is the **mode** of the positive sub-pixel day-of-year values in the block, tested against the region's label window. Be-
1150 cause the exported raster carries no out-of-window positives, in this dataset a single in-window positive sub-pixel makes the cell burned. The fraction of positive sub-pixels agreeing with the modal date is recorded as **burn_date_pixel_agreement_fraction** but no agreement threshold is imposed. The label never affects a cell's eligibility for modelling: unburned, all-no-data
1155 and out-of-window cells all remain as the negative class.

Two safeguards are recorded rather than assumed. Cells that burned inside a region's own predictor window are removed from its analysis universe rather than counted as unburned, which ran for Muğla, Evia and Montiferru, is recorded as not run for Manavgat and has no recorded status for Bejís.
1160 Burning in earlier years is screened for no region in the five-region cohort; the only historical exclusion in the study removes the 2021 Muğla scar from the 2022 event-relative experiment of Section 3.14.

Before any modelling each region passes a gate that answers one question: of the cells labelled burned, what fraction is dominated by natural
1165 vegetation? A region is admitted as a wildfire candidate when that fraction reaches 0.50 and at least 30 burned cells are present, and is rejected as a

cropland-dominated control when the cropland fraction reaches 0.50 instead. The gate uses ESA WorldCover classes aggregated to the same reconstructed cells. Its purpose is to separate burned area produced by natural-fuel combustion from burned area produced by post-harvest stubble burning, which
1170 MCD64A1 does not distinguish. Verdicts are reported in Section 4.1.

Appendix C.2. C.2 Transferability diagnostics versus transfer

Twenty candidate diagnostics from four families are each rank-correlated against the same target quantity, the raw thermal transfer AUC over the
1175 twenty ordered directions, under one common pair-based bootstrap. The families are marginal predictor-distribution measures $P(x)$, burned-niche overlap $P(x|y=1)$, fire-regime label-pattern structure $P(y)$, and conditional feature-response direction $P(y|x)$.

The **marginal** family includes area-of-applicability-style dissimilarity in
1180 scaled, importance-weighted predictor space [21, 22, 16], climatic and geographic distance, and a learned domain classifier trained to separate source from target cells. The **niche-overlap** family includes Schoener’s D [34] and Warren’s I [41] over the burned cells of each region. The **regime** family includes burned-patch component counts and effective component counts. The
1185 **conditional** family is the sign-agreement index: the fraction of predictors whose signed univariate association points the same way in both regions, computed over all predictors and over the subset whose associations are individually interval-supported.

Two properties of this design are stated in advance because they bound
1190 what it can show. The candidate set was fixed in a project analysis log before the correlations were computed; that log is not a formal pre-registration and

no independent timestamped registration exists. And with five regions the effective sample is ten unordered pairs, so nineteen computed variants are evaluated at that power with no family-wise error control claimed; where a
1195 diagnostic clears an interval test but not a Bonferroni threshold, both are reported.

Appendix C.3. C.3 Same-geography event-to-event comparison (Mugla 2021 versus 2022)

Muğla is the one region for which a second fire event is analysed on the
1200 identical AOI and analysis grid, which allows the direction of feature-label associations to be compared with geography held fixed. The 2022 experiment is registered with windows anchored to its own event: a 58-day predictor window closing the day before ignition and a 49-day label window opening on it, matching the 2021 durations exactly.

1205 **This is not a clean temporal-transfer design and is not presented as one.** The 2022 event ignites about six weeks earlier in the season than the 2021 event, so calendar year and seasonal phase are confounded and no difference can be attributed to the year alone. The registered claim is same-geography event-to-event. The pair is deliberately kept out of the twenty-
1210 direction matrix, and that exclusion is enforced in released code rather than merely asserted here.

Two further properties bound the comparison. A historical-burn exclusion applies to the 2022 arm alone, masking every cell that burned in 2021 so the previous year’s scar does not enter the following year’s analysis: it removes
1215 3,073 cells, of which 2,941 belong to the primary population, taking it from 41,730 rows to 38,790, a drop of 7.0 %. The two arms therefore share 38,789 of

38,790 cells and three byte-identical static predictors, and in the 2022-to-2021 direction every target positive lies outside the source training population. Section 4.8 states what follows for how these two numbers may be read.

1220 **CRedit authorship contribution statement**

Yunus Emre Cogurcu: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review and editing, Supervision. **Emrehan Metin:** Software, Data curation, Investigation, Validation, Writing – review and editing.

1225 **Declaration of competing interest**

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All satellite inputs are public and are obtained through Google Earth Engine. The processing pipeline, its configuration and the frozen numeric
1235 outputs are publicly available; the repository, the commit of record and the licence are given in the data and code availability statement of the Methods,

together with the three components that fall outside that release. No digital object identifier is minted and no archival deposit exists.

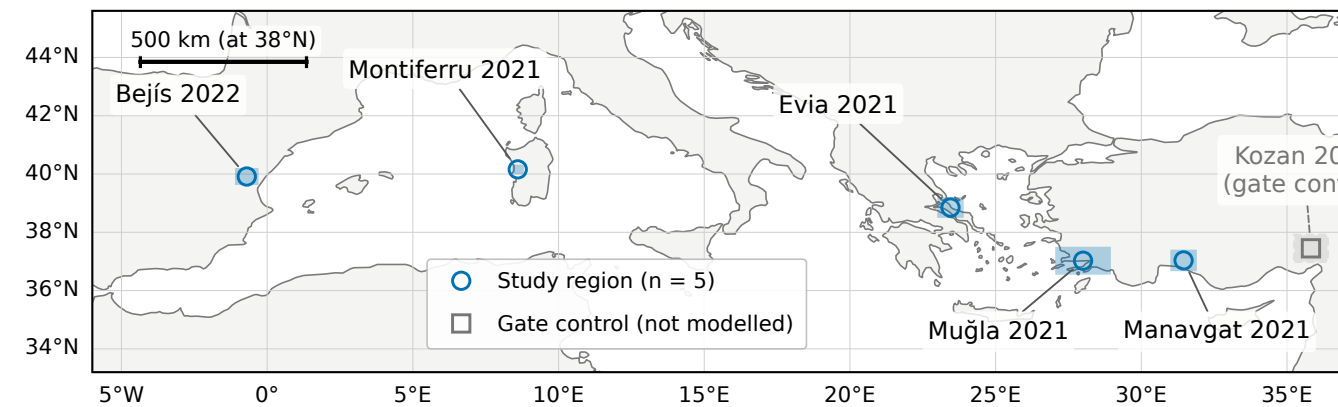


Figure C.1: **Study regions and the gate negative control.** Locations of the five wild-fire regions analysed in this study (blue, circular markers): Manavgat 2021 and Muğla 2021 (Türkiye), Bejís 2022 (Spain), North Evia 2021 (Greece, extended AOI) and Montiferru 2021 (Sardinia, Italy). Rectangles are the true analysis areas of interest; the ring marks each centroid, because at basin scale the rectangles alone are hard to locate. Kozan 2023 (grey, dashed outline, square marker) is *not* a study region: it is the negative control used to calibrate the burned-landcover admissibility gate, and it enters no model, transfer or diagnostic anywhere in this paper. Its rectangle is the true extent of its processed AOI and is larger than the study AOIs for that reason alone; the difference in size carries no methodological meaning and is not a design choice. Coastlines are Natural Earth 1:50 m; country borders are omitted deliberately. The scale bar is exact only at 38°N. Region bounding boxes match Table 1 to 5×10^{-3} degrees.

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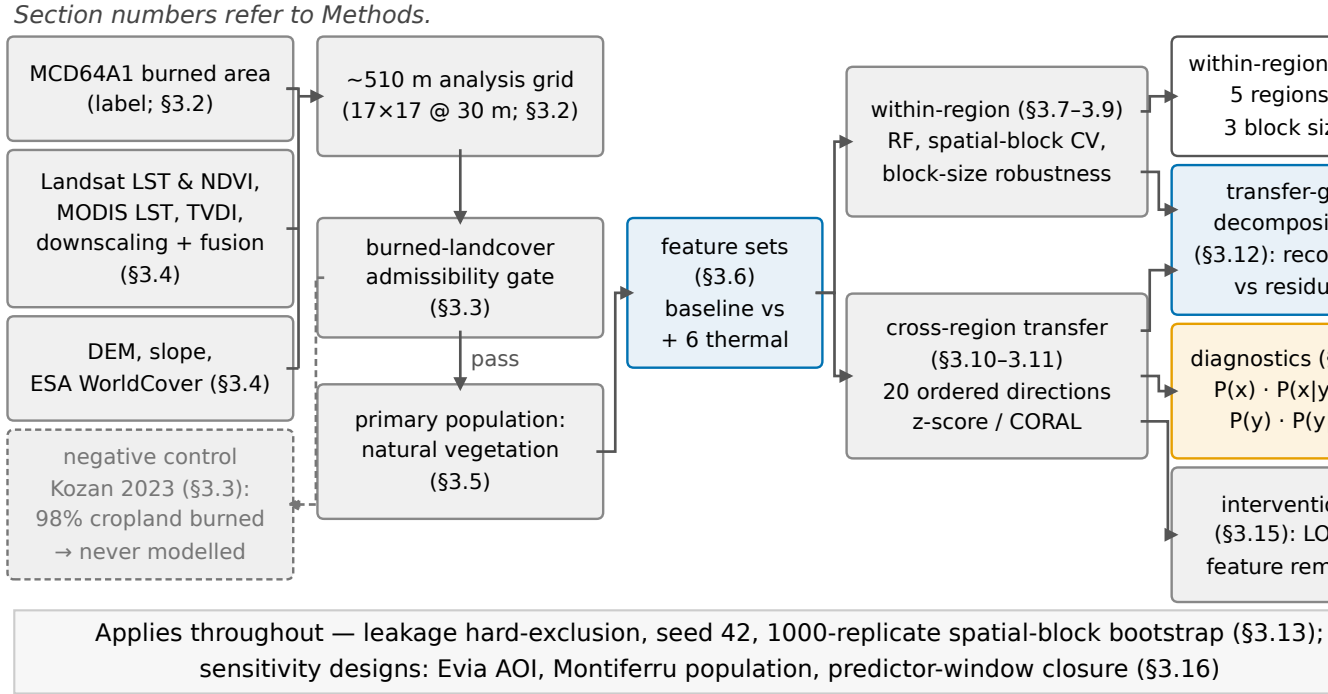


Figure C.2: Processing pipeline and evaluation programme. Flow from source products (left) to the transfer and diagnostic programme (right); every box carries the Methods subsection that specifies it. The five source products (MCD64A1 burned area, Landsat Collection 2 Level-2, MOD11A1, Copernicus DEM GLO-30 and ESA WorldCover) merge onto the reconstructed ~ 510 m analysis grid (17×17 pixels of the 30 m reference grid). The burned-landcover admissibility gate has two outcomes: regions that *pass* proceed to the natural-vegetation primary population, while the rejecting branch (grey, dashed) terminates in the Kozan 2023 negative control, whose burned area is 98% cropland and which is never modelled. The dashed connector deliberately carries no arrowhead onward, so the control cannot be read as entering the pipeline. Feature sets feed two evaluations in parallel: within-region (random forest under spatial-block cross-validation) and cross-region transfer (source-only, 20 ordered directions, with label-blind z-score and CORAL adaptation). Their levels are combined in the transfer-gap decomposition (§3.12). The footer states what applies throughout: leakage hard-exclusion, seed 42, 1000-replicate spatial-block bootstrap (§3.13), and the sensitivity designs (§3.16). Section references are verified against the Methods heading list when the figure is built.

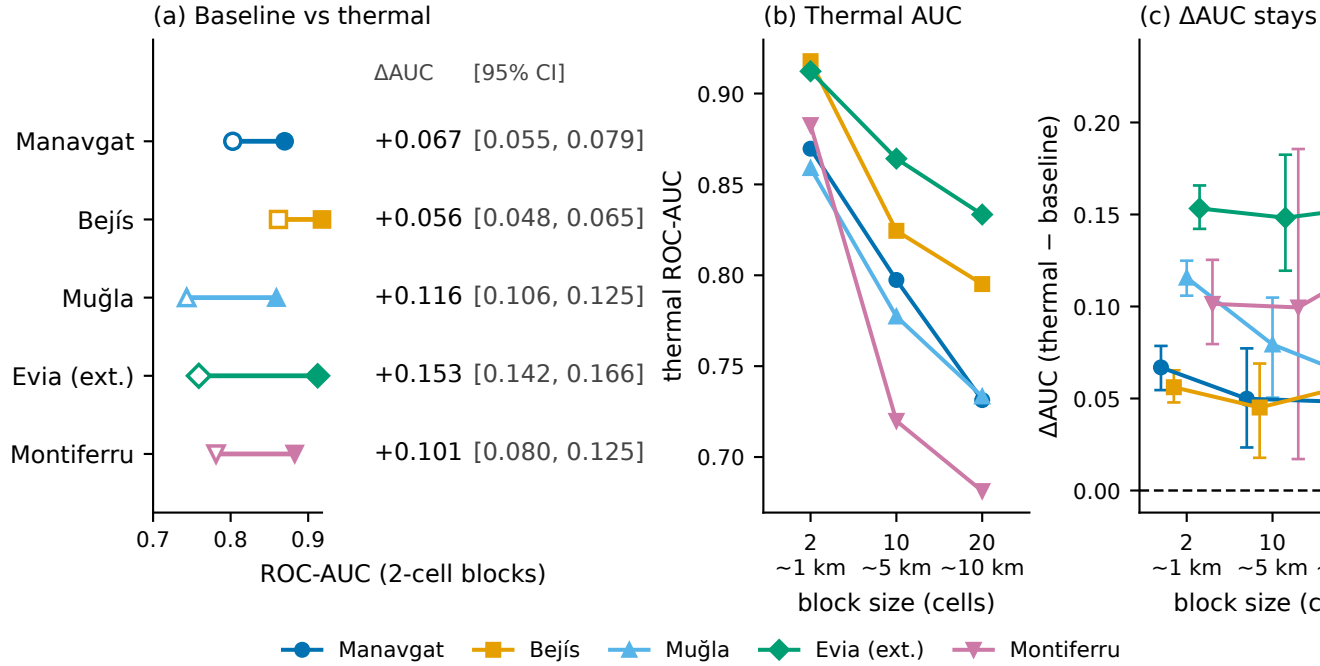


Figure C.3: **The thermal increment replicates in all five regions and survives coarser spatial blocking.** Natural-vegetation primary population; random forest with the primary weighted configuration; all intervals are 95% spatial-block bootstrap intervals (1000 replicates, seed 42). **(a)** Baseline versus thermal out-of-fold ROC-AUC per region at the default 2-cell (~ 1 km) blocking: the open marker is the static baseline feature set, the filled marker is the baseline plus the six thermal predictors, and the connecting bar is the increment. The right-hand column reports the paired ΔAUC and its 95% interval; it is a fixed column placed beyond the plotted AUC range, so no label position depends on a data value. Every interval excludes zero. **(b)** Absolute thermal ROC-AUC against spatial block size (2, 10 and 20 cells, $\approx 1, 5$ and 10 km): absolute skill falls as blocking coarsens, in every region. **(c)** The thermal increment ΔAUC against the same block sizes, with 95% intervals: the increment does *not* melt away, and its interval excludes zero in all ten of the 2- and 10-cell region-block-size combinations. The five 20-cell intervals also lie above zero but rest on 6 to 33 positive-carrying blocks each, so they are plotted as indicative rather than as intervals (Table 2, table note). In (c) the five regions are offset horizontally by 0.15 cell so their whiskers separate; **this offset is cosmetic, since all five regions are evaluated at exactly the same three block sizes** (2, 10, 20 cells), and the offset is smaller than half the spacing between block sizes, so the groups cannot merge. Colour and marker shape both encode region, so the panels remain readable in greyscale.

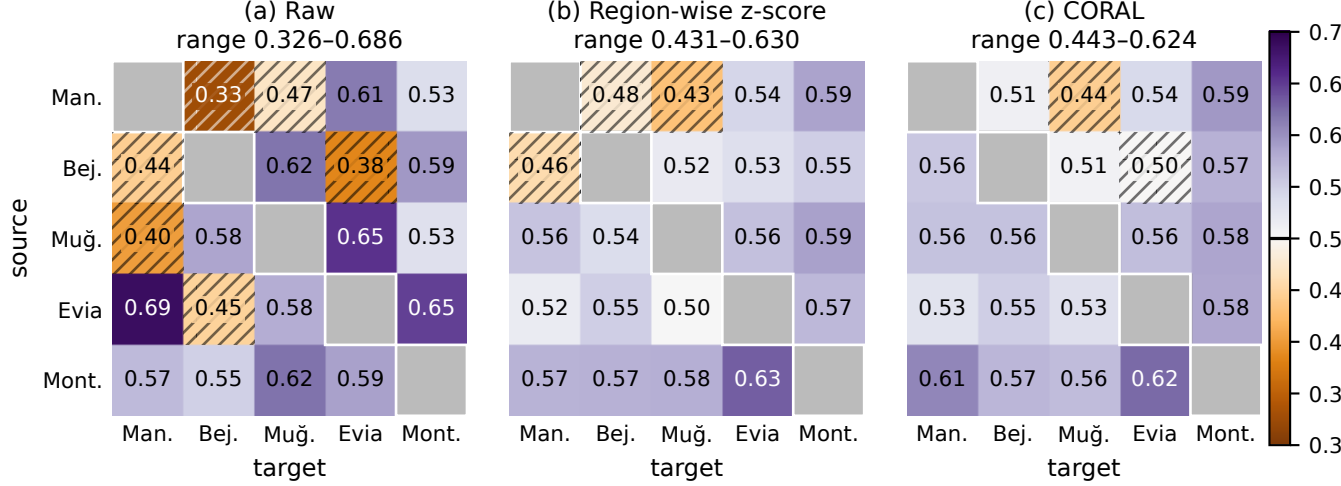


Figure C.4: **Cross-region transfer, and what label-blind adaptation does to it.** Thermal transfer ROC-AUC for all 20 ordered source–target directions among the five regions, natural-vegetation primary population. **(a)** Raw transfer, no adaptation. **(b)** After region-wise z-score standardisation. **(c)** After CORAL. Rows are the source region, columns the target; the grey diagonal is within-region performance and is *not* a transfer result. The diverging colour scale is centred exactly on chance (0.5, marked by the black rule on the colour bar), so orange is below chance and purple above. Each panel header states that panel’s range: adaptation compresses the spread from 0.326–0.686 (raw) to 0.431 to 0.630 (z-score) and 0.443 to 0.624 (CORAL). It pulls the extremes toward chance rather than lifting transfer. Cells below chance are hatched. The hatch is a redundant cue for print: a diverging palette is symmetric in luminance about its centre, so in greyscale a below-chance cell would otherwise be indistinguishable from an equally distant above-chance one. Hatching follows the *exact* value while the printed label is rounded to two decimals, so two cells print “0.50” with different hatching: CORAL Bejís→Evia is 0.4991 (hatched, below chance) and z-score Evia→Muğla is 0.5013 (un-hatched, above chance). Every cell carries its value, and every label’s colour is chosen by measured WCAG contrast against that cell’s rendered colour (worst case 4.7:1). Confidence intervals for these directions are given in Table 4.

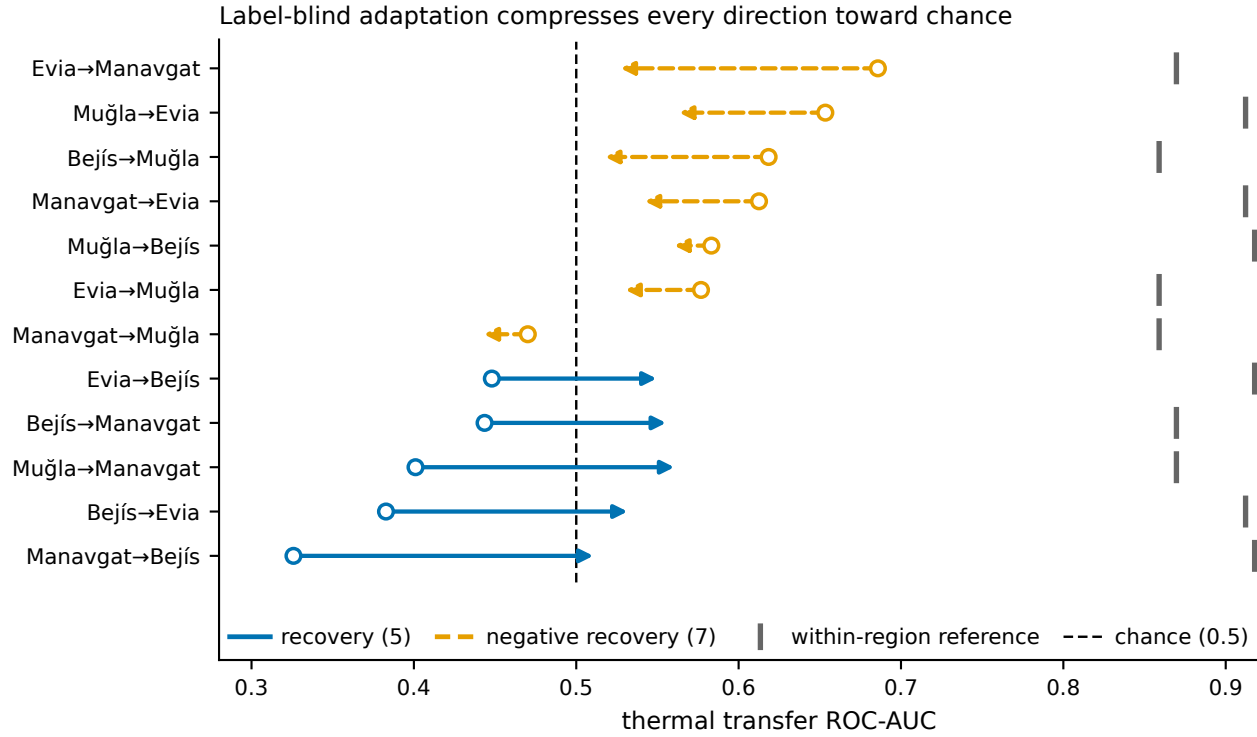


Figure C.5: **Label-blind adaptation compresses transfer toward chance rather than recovering it.** All 12 ordered directions among the four regions carrying the full decomposition (Manavgat, Bejís, Muğla, Evia), natural-vegetation primary population. Each row is one direction. The open circle is raw transfer ROC-AUC and the arrow runs to the best label-blind adapted result for that direction (region-wise z-score or CORAL, whichever is better); the grey vertical tick is the within-region reference for that target, and the dashed vertical rule is chance. **Solid blue arrows are the five directions where adaptation moves transfer toward the within-region reference (recovery); dashed orange arrows are the seven where it moves *away* from it (negative recovery).** Line style and colour both encode this class, because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. The pattern is the point: all six directions starting above chance are pushed down, and five of the six starting below chance are pushed up, so adaptation acts on the spread rather than on skill. The exception is Manavgat→Muğla, which starts below chance at 0.470 and is pushed further down to 0.443. The worst case is Evia→Manavgat, where raw transfer of 0.686 is dragged down to 0.527 against a within-region reference of 0.870, a recovered fraction of -0.862 . No arrow reaches its within-region tick.

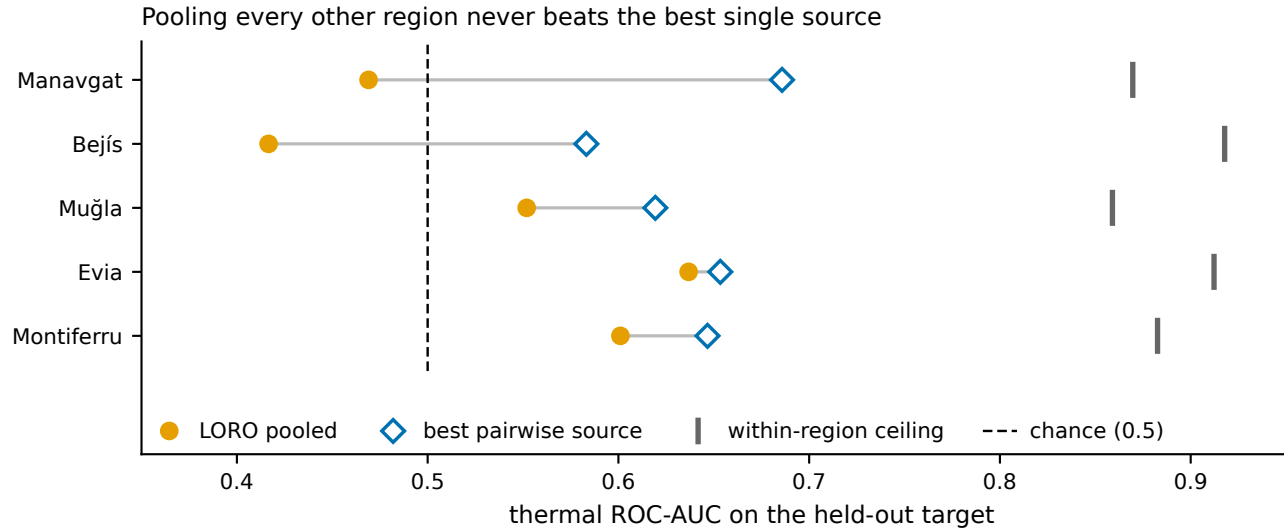


Figure C.6: **Pooling every other region never beats the best single source.** Leave-one-region-out evaluation: for each held-out target region (rows), a model trained on the pooled remaining four regions (filled orange circle) is compared against the best individual source region for that same target (open blue diamond), with the within-region ceiling as a grey vertical tick and chance as the dashed rule. Natural-vegetation primary population, thermal feature set, no adaptation. The pooled model lies to the left of the best pairwise source in all five targets without exception. Adding source regions does not accumulate transferable signal. In two targets the pooled model falls *below chance* at the point estimate (Manavgat 0.469, Bejís 0.417), and only for Bejís does the interval lie entirely below chance, while a single well-chosen source reaches 0.686 and 0.583 respectively. Every point remains far below the within-region ceiling.

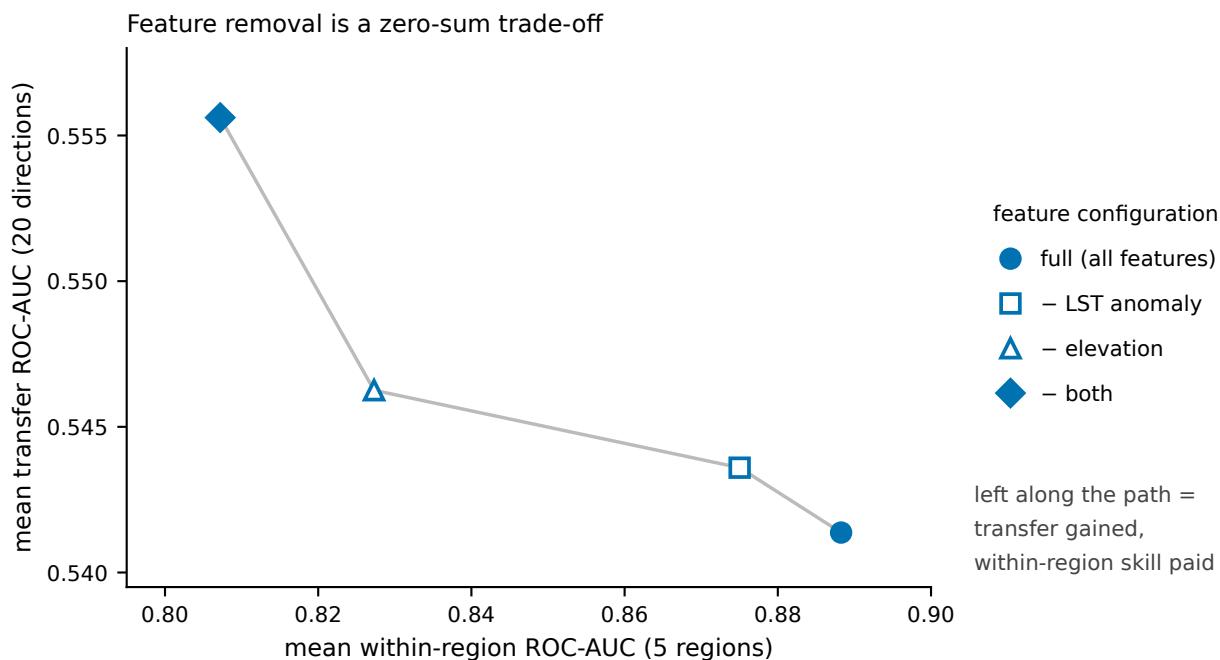


Figure C.7: **Removing the direction-reversing features is a zero-sum trade-off.**

Mean within-region ROC-AUC across the five regions (horizontal) against mean transfer ROC-AUC across the 20 ordered directions (vertical), for four feature configurations: the full thermal set, the set without the LST anomaly, without elevation, and without both. Each configuration has its own marker and is named in the key; the grey path connects them in that order. Moving left along the path, transfer rises monotonically (0.541 $\rightarrow$ 0.556) while within-region skill falls monotonically (0.888 $\rightarrow$ 0.807). Both monotonicities are verified when the figure is built, so the figure cannot silently contradict the claim. The transfer gain is 0.014 AUC and the within-region cost 0.081 AUC, about six times larger: what is removed is locally informative and non-portable, and removing it buys portability by giving up skill rather than by finding transferable signal.

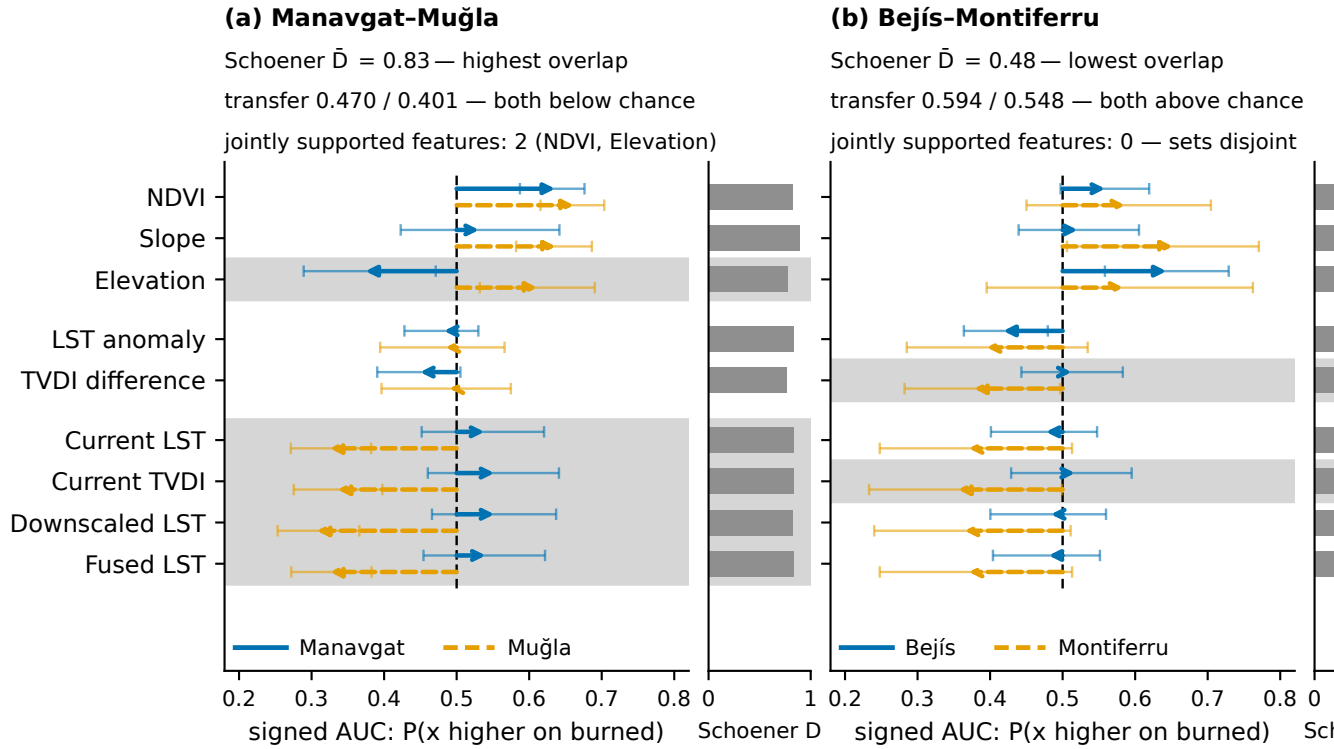


Figure C.8: **Burned-niche overlap does not determine transfer: the contrast pair.** Per-feature signed univariate AUC, the probability that a predictor takes a higher value on a burned cell than on an unburned one, for the two most informative region pairs, natural-vegetation primary population. Arrows run from chance (0.5) to the observed value, so their direction is the sign of the association and their length its strength; whiskers are 95% spatial-block (≈ 5 km) bootstrap intervals. Within each pair the two regions are distinguished by colour, by line style (solid for the first region named, dashed for the second) and by vertical offset; line style is included because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. **A filled arrowhead means that region's 95% interval excludes 0.5, so the direction is statistically supported; an open arrowhead means the interval contains 0.5 and the direction is not supported.** Rows shaded grey are features whose sign disagrees between the two regions. Grey side bars are the per-feature Schoener's D overlap. **(a)** Manavgat–Muğla, the pair with the *highest* burned-niche overlap ($\bar{D} = 0.83$), transfers below chance in both directions (0.470 and 0.401). **(b)** Bejís–Montiferru, the pair with the *lowest* overlap ($\bar{D} = 0.48$), transfers above chance in both directions (0.594 and 0.548). High niche overlap is therefore not sufficient for transfer. The arrowhead fill also exposes the coverage limit of the supported-agreement index, which is defined only on features whose direction is supported in *both* regions: there are two such features in (a) (NDVI and elevation) and **none in (b)**, where the two regions' supported sets are disjoint and Montiferru's intervals

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